

A project on
AUTOMATED POLYHOUSE USING IOT FOR HYDROPONIC
FARMING

SUBMITTED BY

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In partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

Under the guidance of
MR. NISHANT UPADHYAY
Assistant Professor



SHARDA
UNIVERSITY
Beyond Boundaries



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SHARDA SCHOOL OF COMPUTING SCIENCE & ENGINEERING
SHARDA UNIVERSITY, GREATER NOIDA

APRIL, 2026

DECLARATION

I hereby declare that the project work entitled “Automated PolyHouse Using IoT for Hydroponic Farming” submitted to Sharda University, Greater Noida is a record of original work carried out by us under the guidance of Mr. Nishant Upadhyay and this research is being submitted to fulfil the requirements for the award of degree of Bachelor of Technology in Computer Science & Engineering.

The results embodied in this research work have not been submitted to any other university or institution for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the report entitled “Automated PolyHouse Using IoT for Hydroponic Farming” submitted by Mehul Kumar (2022338119), Aryan Raj (2022504700) and Sivam Visnu (2022513718) to Sharda University, to the fulfillment of requirements of the degree of “Bachelor of Technology” is record of Bonafide final year Project work carried by out by them in the “Department of Computer Science & Engineering, Sharda School of Computing Science & Engineering, “Sharda University”.

The results contained in this Project have not been submitted in part or full to any other university or institute for award of any other Degree or fellowship.

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Head of the Department

Department of Computer Science & Engineering

Place:

Signature of External Examiner

Date:

ACKNOWLEDGEMENT

The major project plays an important role in developing practical understanding and applying theoretical knowledge to real-world situations. It provides an opportunity to explore concepts in depth, build technical skills, and understand real-world challenges. We are truly grateful for the opportunity to work on this project, which has contributed significantly to our learning and experience.

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SYMBOLS AND ABBREVIATIONS

AC	Alternating Current
ADC	Analog-to-Digital Converter
AFCI	Arc-Fault Circuit Interrupter
BOM	Bill of Materials
CNN	Convolutional Neural Network
CPSC	Consumer Product Safety Commission
CSV	Comma-Separated Values
DWC	Deep Water Culture
EC	Electrical Conductivity
ELCI	Equipment Leakage Circuit Interrupter
ESP32	Microcontroller / Development Board
ESFI	Electrical Safety Foundation International
GFCI	Ground-Fault Circuit Interrupter
GFPE	Ground-Fault Protection of Equipment
HOD	Head of Department
HUD	U.S. Department of Housing and Urban Development
IoT	Internet of Things
KPI	Key Performance Indicator
LDR	Light-Dependent Resistor
MCB	Miniature Circuit Breaker
MCU	Microcontroller Unit
NEC	National Electrical Code
NFT	Nutrient Film Technique
NIST	National Institute of Standards and Technology
OSHA	Occupational Safety and Health Administration
OTA	Over-The-Air
PID	Proportional-Integral-Derivative
PSU	Power Supply Unit
RH	Relative Humidity
ROI	Return on Investment

SoC	System-on-Chip
SOP	Standard Operating Procedure
SSR	Solid State Relay
TDS	Total Dissolved Solids
UART	Universal Asynchronous Receiver-Transmitter
PSU	Power Supply Unit
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SoC	System-on-Chip
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CHAPTER 1:

INTRODUCTION

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INTRODUCTION

Rapid urbanization and water scarcity are pressuring food systems to produce more with fewer resources. This project develops a low cost, open, and modular automation framework for a vertical hydroponic tower suitable for small urban polyhouses. The system integrates an ESP32 S3 controller, recirculating irrigation, and multi sensor monitoring (pH, EC/TDS, solution temperature, ambient temperature/humidity, and level) with a sequential dual loop strategy that corrects pH first and then nutrient concentration in small, timed doses. A mobile application provides real time telemetry, basic actions, and lightweight analytics, while safety features—GFCI protection, per channel fusing, leak/level cut offs, and store and forward logging—support reliable operation in constrained indoor settings. The prototype emphasizes affordability by using commodity pumps, budget pH/EC front ends with disciplined calibration, and a minimal bill of materials under a student budget. Bench and short grow trials on leafy greens demonstrate stable solution control within practical bands (e.g., pH 5.8–6.2, EC tailored to crop stage), smooth 8–24 h unattended runs, and clear paths to water and energy savings via recirculation and scheduled lighting. Furthermore, the current phase of this project upgrades the framework from passive monitoring to an active Cyber-Physical System. This Phase 2 integration introduces an Artificial Intelligence (AI) layer utilizing an ESP32-CAM vision node to run lightweight Convolutional Neural Networks (CNNs) for real-time plant disease detection. Concurrently, the dual-loop control logic is physically realized using an array of 12V peristaltic pumps, enabling autonomous, high-precision nutrient and pH dosing without human intervention. This work shows that dependable, data driven hydroponic automation can be achieved at low cost, providing a scalable template for urban food production and classroom labs.

INTRODUCTION AND OVERVIEW

1.1.1 Motivation

The world's urbanization is accelerating at a record pace, while opportunities for farming using conventional agriculture methods (i.e., soil-based) are decreasing at an alarming rate. This disparity is particularly evident in densely populated urban areas where residential, commercial and industrial uses compete directly with the production of food. In addition, as urban dwellers show a growing desire to buy

locally grown and pesticide-free leafy greens produced as close to them as possible, this situation has presented cities, institutions and consumers alike with the need to look into multiple compact, indoor and semi-controlled growing systems (that can operate on a year-round basis) that can produce food with minimal land and water resources. Hydroponics has quickly gained popularity among all food production systems, because it provides the greatest water savings, produces crops at a faster rate than other systems, and is ideally suited for use in both indoors and vertical farming.

While there are many advantages associated with hydroponic systems, adopting them outside of a commercial operation remains rare for most small-scale growers. Typically, there are two major obstacles that small-scale structures face when they try to grow using hydroponics. First, the high cost of commercial hydroponic towers and automated grow systems often makes them impractical for small-scale growers to use. Because many of these products are proprietary and have a closed system (i.e., the components cannot be modified or repaired), they become difficult for growers to maintain, document, and study for educational purposes (Holmes et al., 2022). Second, while there are numerous examples online of low-cost DIY systems, many of them are poorly constructed and unsafe. Without stable construction, these systems often provide little documentation or assurance of stability, making them difficult to maintain (Holmes et al., 2022). Most of these DIY systems have many of their critical components (such as pH and electric conductivity of the nutrient solution) left up to the grower to adjust by hand; thus, they are not able to maintain stable pH and electrical conductivity, which will drift over time. The pH of the nutrient solution drifted up (Holmes et al., 2022).

Unstable pH and nutrient concentrations are among the most common reasons for the failure of small hydroponics systems. If the pH is not within the correct range for optimal growth, then nutrients will not be available to the plant. The same goes for if a grower's EC (Electrical Conductivity) is not at the proper level because it will cause deficiency of nutrients and excessive salt build-up in the system can damage the roots. All of these can have a direct effect on the growth of their plants with poor growth, leaf yellowing, root rot/disease, as well as clogging pumps and delivery lines. In addition, because new users will have to frequently correct their systems manually due to these instabilities it creates much more work for the operator throughout the

course of the day and a lot of resources (water, nutrients, and time) end up wasted. Additionally, for most new users the continual failures resulting from these instabilities are very discouraging and will probably lead to the complete abandonment of the hydroponics system.

This initiative aims to fill the ongoing need for an affordable, repeatable automated system, using an hydroponic vertical tower in urban areas including polyhouses, houses and schools in the future through creating and implementing a documentation of a process that automates the monitoring and feeding of nutrient solutions within prescribed ranges of safe and crop spec through incorporated automation to minimize the amount of human time required to manually operate. An established checklist of daily actions and procedures for installations, calibration, and preventative health care supports reducing the number of first-time users and increasing system reliability over time.

Substantiating it further, the focus of this project is not simply to show off a functional proof-of-concept. Rather, it is to create an inexpensive, sustainable, reliable and user-friendly (i.e., "unattended") automated hydroponic system that can provide nutrient and hydration services for extended periods of time without maintenance; and, in the event of a temporary loss of electricity or network connectivity, to restore its previous state without operator involvement. Data logging of operational metrics and transparent control logic also allow for the collection of historical performance data to enable further analysis, optimization, and improvement of the system over time. As such, the goal of this initiative is to create an opportunity for people to be able to use automated hydroponics as an affordable, stable, and safe method of growing food in urban environments for smaller-scale operations.

1.1.2 Overview

A vertical NFT hydroponic tower has been created. The Hydroponic tower system consists of a light-tight nutrient reservoir. An immersed electric pump moves the nutrient solutions up and down the Hydroponic tower. As the nutrient solutions flow down through the gently sloped channels to the plants, a thin layer of nutrient solution is formed.

Once the nutrient solutions reach the plant roots, they are returned to the reservoir. The use of an NFT configuration to create a recirculating hydroponic growing system allows for improved water conservation, even nutrient delivery and a compact vertical growing approach, which makes it ideal for small urban buildings or indoor growing environments.

The full Hydroponic tower system is managed by an ESP32-S3 microcontroller. It collects both nutrient and environmental data from various sensors. pH levels, Electrical Conductivity (EC) and Total Dissolved Solids (TDS) Of the Nutritional Solution, Nutritional Solution Temperature, Ambient Temperature and Humidity levels, and the amount of water Remaining in The Hydroponic Tower are monitored using the above sensors.

After collecting and analyzing the input, the microcontroller will then deliver commands to all of the pumps, lights and other devices connected to the tower. All devices (such as pumps, lights, etc.) are connected to the microcontroller via electrically isolated relays or solid-state relays (SSRs). The SSRs allow for safe separation between the low voltage control electronics and the high voltage power source.

A Dual Loop System Control approach is used to control the pH of the nutrient solutions and their electrical conductivity (EC). Loop One will provide corrections for the pH of the nutrient source by administering a small corrective dosage of the pH solution of choice (either up or down) into the nutrient solution, waiting for the appropriate amount of time to mix and settle before checking the pH to see if it falls within the desired limit band. Once the pH is within the desired band, Loop Two will adjust the nutrient strength by administering EC corrections to meet the target EC value (crop specific). This serial method eliminates possible chemical interactions, decreases overshooting, and enhances nutrient solution stability compared to concurrent correction.

Using Flutter, a lightweight mobile app provides user access to functionality. The app displays a real-time dashboard displaying essential nutrient solution parameters such as pH, EC, temperature, humidity, and tank water level, as well as the ability to manually start irrigation/fertilization through a quick access notification if any

parameter exceeds acceptable limits. A straightforward historical view can help users quickly assess trends without needing extensive data analysis.

A significant point of emphasis in the design is the size limitations/constraints for both Student Lab and classroom Use. To promote low-cost reproduction and to provide for a more consistent result, commodity Materials have been selected for construction of all the components of this system. All Electrical Wiring is neatly kept together and labeled for easy identification; also included in the system is GFCI Protection & fusing that ensures the safety of the entire system during operation. Sensor Probes are enclosed within protective sleeves and utilize sealed cable Glands. Through the use of clear labeling and an intuitive interface, all personnel on the team can easily operate, maintain, and service the system and will be able to quickly identify any subsequent issues.

Data associated with Operations will be stored as Compact JSON Events, which will then be periodically transferred to CSV Format for ease of analysis in common Spreadsheets. This means that at any time that Event Data needs to be reviewed, the results logged against the Event can be analyzed very efficiently. Ultimately, the integration of Standard Engineering Practices (Safe electrical and plumbing design) and Control Logic, along with the provision of Telemetry, demonstrate the system's stability and reliability, as well as its potential savings in terms of operational resources.

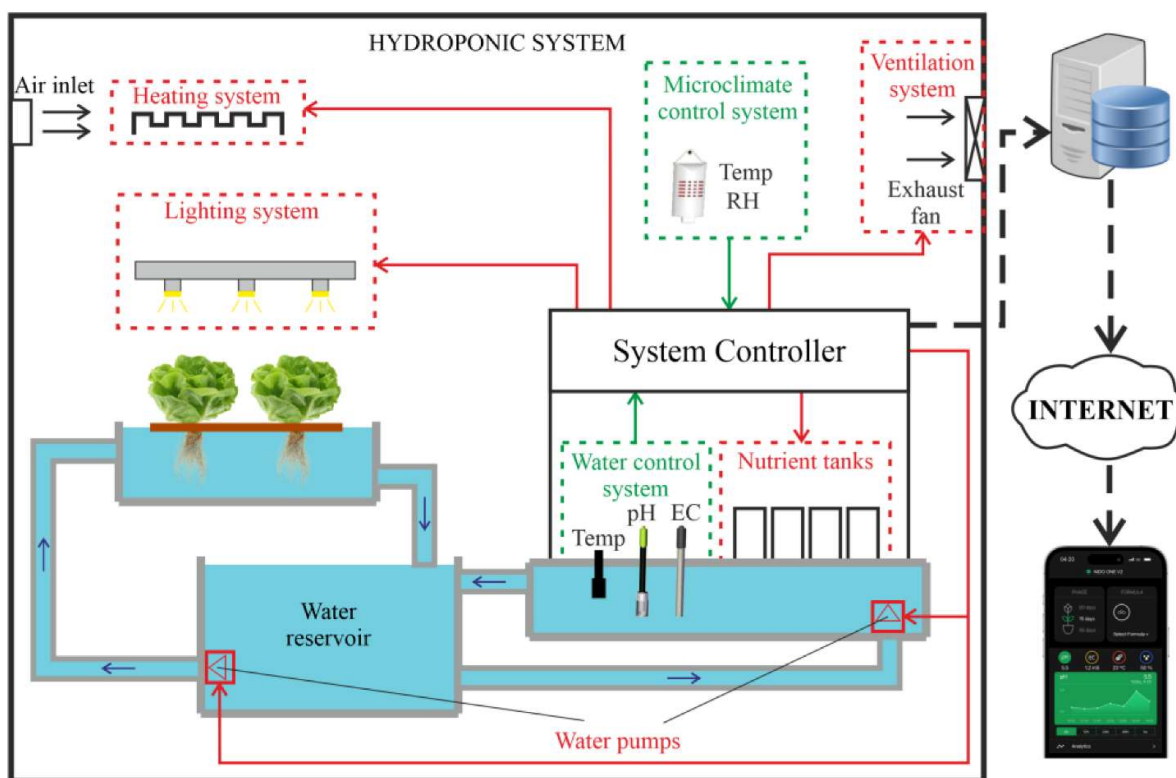


Figure 1.1: Diagram of a Hydroponic System

1.1.3 Expected Outcome

Operational prototype: Vertical tower NFT operation capable of running unattended for 8–24 hours, with the ability to maintain crop-appropriate nutrient solution chemistry within target pH and EC levels for much of the day (e.g., pH range of 5.8–6.2 and stage-dependent EC).

Traceable data logs: Multi-Day Trials Generate Time-Stamped Recordings of pH, EC, Solution Temperature, Ambient Temperature/Humidity, Water Level and Dosing. Multi-day trials generate time-stamped recordings of pH, EC, solution temperature, ambient temperature/humidity, water level and dosing events (suitable for the generation of stability, recovery following corrective actions and gradual drift plots).

Monitoring application: A mobile app that provides users with real-time access to the sensor data, as well as the ability to set up alerts for when thresholds are exceeded, is an excellent way to facilitate communication and track activity (including manual dosing and maintenance).

Build and operation documentation: Comprehensive documentation, including the priority-based BOM, electrical wiring diagrams with safety information, pH and EC calibration procedures, and daily as well as weekly checklists.

Validated trial results: The results of the growth and testing have shown that when continuous delivery is provided at a controlled and limited rate (i.e., no overshoot), much less labor will be required due to labor savings from using the recirculating delivery method, and the evaporation losses incurred from this delivery method will also be reduced, thus providing for the potential scalability of this process to other crops and towers.

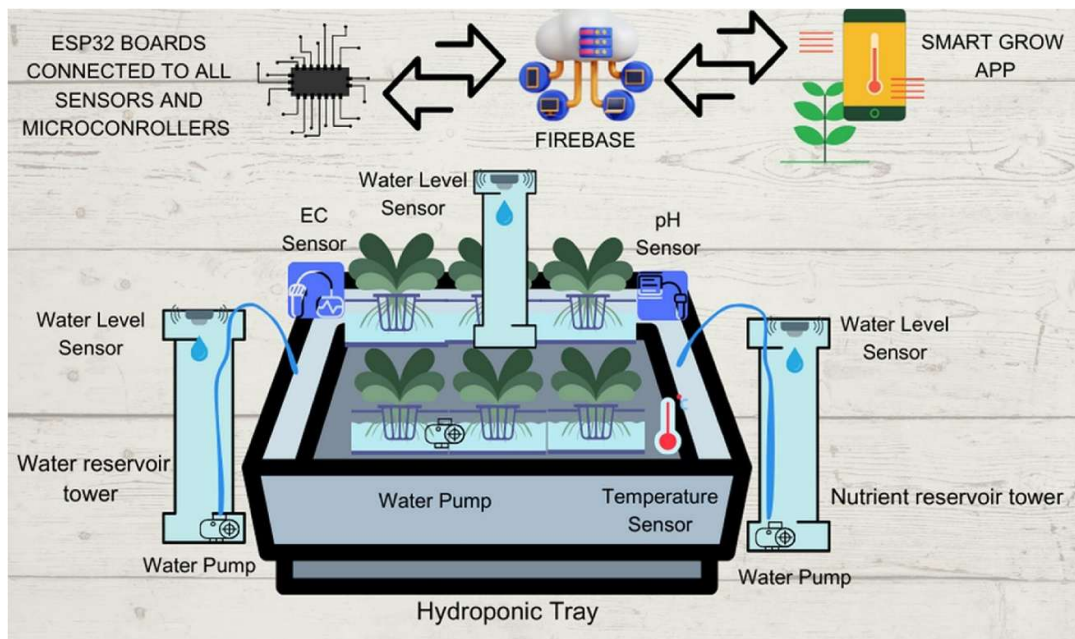


Figure 1.2 : Smart Grow App System Architecture

1.1.4 Gantt Chart

The main project tasks are shown on this Gantt chart, along with their start and finish dates and weekly duration. It guarantees that tasks are finished on time to satisfy project deadlines and aids in the visualization of the project schedule. With the discussion to our respected guide, Mr. Nishant Upadhyay, we have outlined the main tasks assigned to do and has been illustrated below.

Table 1.1: Gantt Chart

Table 1.1: Gantt Chart

S. No	Task / Activity	Start Date	End Date	Duration
1	Literature Review & Hydroponic Systems Study	18 Mar 2025	28 Mar 2025	~10 days
2	Study of Existing Automated Ag-Tech & IoT Domain	28 Mar 2025	09 Apr 2025	~12 days
3	Project Scoping, Hardware Specs & Tower Design	09 Apr 2025	16 Apr 2025	~7 days
4	Plant Disease Dataset Study & Feasibility Analysis	16 Apr 2025	23 Apr 2025	~7 days
5	Sensors & Actuators Research (TDS, pH, L298N, ESP32)	23 Apr 2025	29 May 2025	~36 days
6	AI Vision Model Planning & Architecture (TFLite/MobileNet)	29 May 2025	25 Jul 2025	~2 months
7	Initial Dashboard Visualization (Streamlit + MQTT Setup)	25 Jul 2025	01 Aug 2025	~7 days
8	Basic Edge Connectivity (ESP32 to RPi Communication)	01 Aug 2025	08 Aug 2025	~7 days
9	IoT Monitoring Module Completion (Water & Temp Data)	08 Aug 2025	14 Aug 2025	~6 days
10	Research Paper Drafting (Automated Hydroponic Phase 1)	14 Aug 2025	25 Aug 2025	~11 days
11	Paper Submission & Peer Review Phase	25 Aug 2025	08 Sep 2025	~14 days
12	Training AI Model for Disease & Nutrient Deficiency	08 Sep 2025	22 Sep 2025	~14 days
13	Dataset Enhancement (Custom Leaf Images + Augmentation)	22 Sep 2025	06 Oct 2025	~14 days
14	AI Inference Integration on Raspberry Pi	06 Oct 2025	13 Oct 2025	~7 days
15	Refining Automation Logic (Fan & Pump Control Loop)	13 Oct 2025	27 Oct 2025	~14 days
16	Full System Integration (Nutrient AI +	12 Nov	02 Dec	~20 days

	Water Management)	2025	2025	
17	Hardware-Software Stress Testing & Tower Stability	02 Dec 2025	09 Dec 2025	~7 days
18	Project Report Writing (Initial Draft & Tech Specs)	09 Dec 2025	16 Dec 2025	~7 days
19	Report Completion & Final Prototype Demo Prep	16 Dec 2025	23 Dec 2025	~7 days
20	Final Results Compilation, Validation & Accuracy Tests	09 Mar 2026	23 Mar 2026	~14 days
21	Final Report Writing Phase & Project Presentation	23 Mar 2026	11 Apr 2026	~19 days

Project Diary to be maintained by each student

Project Diary (Student)

Name: Mehul Kumar Roll No.: 2201010402

Project Title: Automated Polyhouse using IoT for Hydroponic Farming

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
One	18/3/25	-	-	Basic study of Hydroponics	<u>[Signature]</u>
Two	28/3/25	Done	Acquired knowledge of Hydroponic Farming	Components List	<u>[Signature]</u>
Three	9/4/25	Done	Formation of component list	Design the framework	<u>[Signature]</u>
Four	16/4/25	Not Complete	Some detailing in design	Read & Research papers	<u>[Signature]</u>
Five	23/4/25	Done	Got some more details	Drafting of research paper	<u>[Signature]</u>
Six	29/4/25	Done	Done with title and abstract	Control Diagram & Relations	<u>[Signature]</u>
Seven	25/7/25	Not Complete	Only related work done	Determination process	<u>[Signature]</u>
Eight	1/8/25	Done	Started with germination	Complete the research paper	<u>[Signature]</u>

Student is required to maintain one separate page for each visit on which he has to write description of the task and brief idea how that task was carried out.

Figure 1.3(a): Figure showing the visits made, representing physical Gantt Chart

Project Diary to be maintained by each student

Nine	8/8/25	Not complete	Calibrate sensors	Improve on mistakes	<i>Aut</i>
Ten	14/8/25	Done	Paper mailed mailed	Assemble DIY environment	<i>Aut</i>
Eleven	25/8/25	Not Done	Assembly incomplete	Check for conferences	<i>Aut</i>
Twelve	10/9/25	Done	Basic structure completed	Add more features	<i>Aut</i>
Thirteen	17/9/25	Not done	check wiring for lapses	Paper title for 2nd paper	<i>Aut</i>
Fourteen	24/9/25	Done	2nd paper done and paper title done	Submission of 2nd paper	<i>Aut</i>
Fifteen	9/10/25	Done	Calibrate SOP	Calibrate SOP	<i>Aut</i>
Sixteen	16/10/25	Not Done	Early draft of 2nd research	Final time sensor placement	<i>Aut</i>
Seventeen	23/10/25	Done	Paper Accepted	Registration for conference	<i>Aut</i>
Eighteen	9/11/25	Done	Beginnings complete	Integration of hardware modules	<i>Aut</i>
Nineteen	12/12/25	Not done	Modules identified	Finalised testing of two hours	<i>Aut</i>

Student is required to maintain one separate page for each visit on which he has to write description of the task and brief idea how that task was carried out.

Figure 1.3(b): Figure showing the visits made, representing physical Gantt Chart

1.1.5 Possible risks

While the goal of our study is to develop a reliable, low-cost automated hydroponic system using IoT and AI-based monitoring, several practical risks and system-level constraints must be critically evaluated to ensure long-term stability, safety, and scalability of the solution.

- i. **Sensor Drift and Calibration Errors:** One of the most critical risks in hydroponic systems is the gradual drift of pH and EC sensors over time. Inaccurate readings can lead to incorrect dosing decisions, resulting in nutrient imbalance or plant damage. This is especially problematic in low-cost sensors used in student-level systems. To mitigate this, periodic calibration schedules, validation against reference solutions, and error threshold alerts will be implemented.
- ii. **Electrical Noise and Signal Interference (EMI):** The presence of AC-powered components such as pumps and grow lights can introduce electromagnetic interference, which may distort analog sensor readings. This leads to unstable control behaviour and oscillations in dosing logic. Proper shielding of signal cables, physical separation between power and signal lines, and use of digital

communication protocols (I2C/UART) are adopted to reduce this risk.

- iii. Pump Failure and Fluid System Blockages: Clogging due to nutrient residues, root debris, or airlocks in tubing can disrupt nutrient flow or dosing accuracy. Additionally, failure of peristaltic pumps can halt automated correction entirely. Preventive measures such as inline filters, routine flushing, and monitoring flow consistency are incorporated to ensure uninterrupted operation.
- iv. Overdosing and Control Logic Instability: Improper tuning of the dual-loop control system or rapid successive dosing without adequate mixing time can lead to overshooting of pH or EC levels. This may create chemical instability or harm plant roots. The system mitigates this through a “dose–mix–wait–measure” strategy, strict dosing limits, and fallback alerts in case of no-response conditions.
- v. Power Failure and System Downtime: Unexpected power outages can interrupt control operations, potentially leading to pump failure, stagnation of nutrient solution, or data loss. Although the system includes watchdog timers and recovery mechanisms, prolonged outages remain a risk. Future improvements may include battery backup or solar integration to enhance resilience.
- vi. Water Leakage and Electrical Safety Hazards: Since the system operates in a water-rich environment with electrical components, leakage or improper insulation can pose serious safety risks. Despite the inclusion of GFCI protection, fuses, and sealed enclosures, improper installation or wear over time can still lead to hazards. Regular inspection and adherence to safety protocols are essential.
- vii. Scalability Constraints: While the current system performs efficiently at a small scale (single tower), scaling to multiple towers or larger farms may introduce challenges such as increased load on the controller, communication delays, and more complex nutrient management. Future architecture upgrades (e.g., LoRa WAN or distributed control nodes) will be required to address this limitation.

1.1.6 SRS

We have categorized the SRS (System Requirement Specification) into 5 categories including Introduction, Functional and Non-Functional requirements, Constraints and Assumptions and Dependencies. They have been briefly explained below:

1.1.6.1. Introduction

1.1.6.1.1 Purpose: This document aims to define the functional and non-functional requirements for designing and developing an IoT-based automated hydroponic system capable of monitoring and controlling environmental and nutrient parameters (such as pH, EC, temperature, and water level) to ensure optimal plant growth with minimal human intervention.

1.1.6.1.2 Scope: The scope of this project includes the design and implementation of a vertical hydroponic system integrated with an ESP32-based controller, multiple sensors, automated dosing mechanisms, and a mobile application. The system will monitor real-time parameters, apply control logic (dual-loop dosing), log data, and provide user interaction through a dashboard for monitoring and manual control.

1.1.6.2. Functional Requirements

1.1.6.2.1 Data Acquisition and Monitoring:

- i. The system will collect real-time data from sensors including pH, EC/TDS, temperature, humidity, and water level.
- ii. The system will continuously monitor and update sensor readings at predefined intervals.
- iii. The system will detect abnormal values and trigger alerts when parameters exceed safe thresholds.

1.1.6.2.2 Control and Automation:

- i. The system will implement a dual-loop control mechanism where pH is adjusted first, followed by EC adjustment.
- ii. The system will activate peristaltic pumps for controlled dosing of nutrients and pH correction solutions.
- iii. The system will follow a “dose–mix–wait–measure” sequence to avoid overshooting and ensure stability.
- iv. The system will automatically control actuators such as water pumps, lights, and optional fans using relays or SSRs.

1.1.6.2.3 Data Logging and Processing:

- i. The system will log sensor readings, dosing events, and system alerts with timestamps.
- ii. The data will be stored locally in JSON format and periodically converted to CSV for analysis.
- iii. The system will support historical data visualization for performance tracking and analysis.

1.1.6.2.4 User Interface (Mobile Application):

- i. The system will provide a mobile application dashboard displaying real-time sensor data.
- ii. Users will receive alerts and notifications when parameters go out of range.
- iii. The system will allow manual override controls such as irrigation and dosing actions.
- iv. The application will display historical trends for better decision-making.

1.1.6.2.5 Safety and Fault Handling:

- i. The system will automatically shut down the pump if the water level is low to prevent dry run.
- ii. The system will enforce maximum dosing limits per hour and per day.
- iii. The system will detect sensor faults and generate appropriate alerts.
- iv. The system will recover automatically after power or network failure using watchdog mechanism.

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- iii. The system will detect sensor faults and generate appropriate alerts.
- iv. The system will recover automatically after power or network failure using watchdog mechanisms.

1.1.6.3. Non-Functional Requirements

1.1.6.3.1 Performance:

- i. The system must maintain pH within the optimal range (e.g., 5.8–6.2) and EC within crop-specific limits.
- ii. The system should support continuous operation for 8–24 hours without manual intervention.
- iii. Data updates and alerts should occur in near real-time..

1.1.6.3.2 Reliability:

- i. The system must operate reliably under varying environmental conditions.
- ii. It should achieve at least 98% uptime during scheduled operation.
- iii. The system must recover automatically after power or connectivity interruptions.

1.1.6.3.3 Security:

- i. The system must ensure secure communication between device and mobile

application.

- ii. Data integrity must be maintained during storage and transmission.
- iii. Unauthorized access to system controls must be restricted.

1.1.6.3.4 Scalability:

- i. The system design should support expansion to multiple hydroponic towers.
- ii. The system should handle increased sensor data and multiple control nodes efficiently.
- iii. Future integration with cloud platforms and IoT networks (e.g., LoRaWAN) should be supported.

1.1.6.4. Constraints

The system must operate within a limited student budget using cost-effective components. It must ensure electrical safety using GFCI protection, proper isolation, and fuses. The design must rely on readily available hardware components and open-source software tools. Environmental constraints such as indoor space, water availability, and power supply must also be considered.

1.1.6.5. Assumptions and Dependencies

The system assumes availability of stable power supply and basic internet connectivity for monitoring. It depends on hardware components such as ESP32 microcontroller, sensors (pH, EC, temperature), relays, and pumps. Software dependencies include embedded firmware, mobile application frameworks, and optional machine learning frameworks for AI-based disease detection.

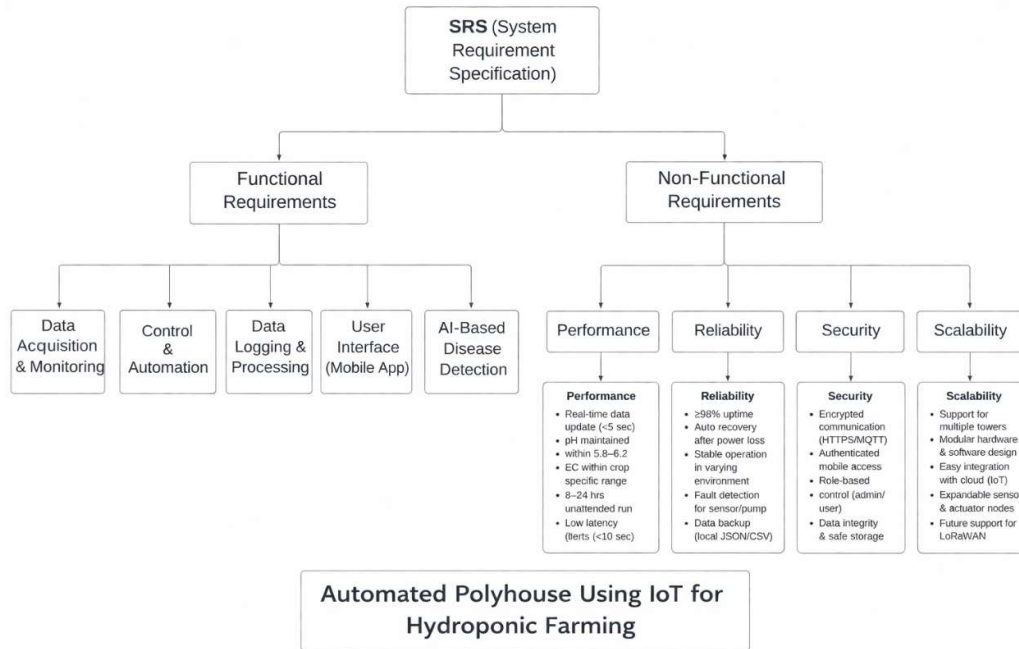


Figure 1.4: System Requirement Specification (SRS)

1.1.7 Hardware & Software Specifications

Hardware

Hydraulics and structure: The vertical NFT tower is the physical planting structure. This structure is fed from a light-proof water reservoir that holds approximately 20 to 40 L of nutrient solution. The water from this reservoir is then pushed through a submersible pump into the NFT tower, creating a continuous film of nutrient solution over the root zone without causing any turbulence in the solution. The nutrient solution is returned to the reservoir via return pipes that are clamped onto the pump for ease of maintenance, and a mesh strainer/filter is installed at the inlet port of the pump to keep root debris and/or sediment from reaching the pump and causing blockages.

The sensing layer is designed to monitor the major, or critical parameters that impact the health of a growing plant through a digital interface connected to a pH probe, a built-in temperature compensated EC/TDS sensor, and a DS18B20 temperature probe

to accurately measure the temperature of the solution. To understand the environmental parameters impacting growth, ambient temperature and relative humidity are measured in the system. A water level sensor (float, optical or pressure based) is integrated into the system to prevent the pump from running dry-sensing and as a safety interlock. For additional diagnostics and system performance Monitoring a flow sensor can also be added to the system.

In regard to an ***Actuation Layer***, small peristaltic pumps dedicated to chemical as well as nutrient control are used to supply the correct pH levels and deliver the correct amount of nutrients into the system. The chemical-resistant tubing along with inline check valves protect against backflow or contamination of chemicals and nutrients during dispensing by the peristaltic pumps, thus allowing accurate dosing. Relay-controlled or SSR outputs are utilized to activate both the main circulation pump and grow lights; additionally, an optional fan output may be added for basic temperature/airflow management around the base of the system (where the nutrient solution is dispersed into the plant tower).

Control / Power Subsystem - The ESP32-S3 development board acts as the central controller and interfaces with sensors, actuators, and mobile app. Power is supplied by a panel-mounted DC power supply (12 / 24 V) with a regulated 5 V rail for logic. The actuators are powered by Optically Isolated Relays (or SSR) providing Electrical Isolation. All components are mounted in an enclosure with Cable Glands, Drip Loops, Grounding, Clear Naming for Terminal Blocks, and Mains Input Protected via GFCI (Ground Fault Circuit Interrupter) and appropriately rated Fuses providing Safe Operation.

Sensors Used:

1. Temperature & Humidity Sensor

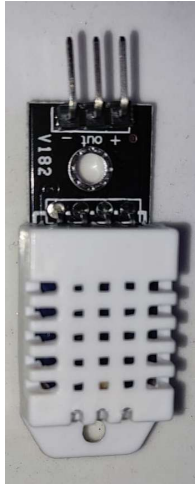


Figure 1.5: Temperature & Humidity Sensor

2. ESP32 Sensor



Figure 1.6: ESP32 Sensor

3. Relay Module



Figure 1.7: Relay Module

4. LDR Module

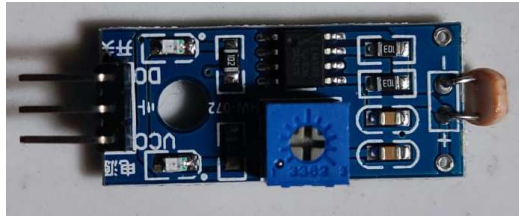


Figure 1.8: LDR Module

5. Water Level



Figure 1.9: Water Level

Software

- **Firmware:** The ESP32-S3 operates with modular firmware, made up of I2C and UART sensor driver modules that work together through a dual-loop control scheme. The first control loop corrects the pH level of the solution, while the second loop manages the conductivity level of the solution. The timing of the respective loops follows a control dose → mix → wait → measure system. The built-in safety features include watchdog timers for automatic recovery after power failures, maximum hourly and daily dosing limits, and a graceful recovery (restart) capability in case of an unexpected system reset. All data collected (measurements and events) are stored locally on the device using a store-and-forward method, which allows for data integrity even when the device loses its network connection, while also providing OTA (Over-the-Air) updates for firmware updates without having to disassemble the device itself.

- **Mobile application:** This Flutter app, which is lightweight, allows for the creation of a real time dashboard that contains live tiles of different data types (i.e. pH, EC, Solution temperature, Ambient temperature/humidity, and Water level). It also includes alert messages to inform users when the system has gone out of range, as well as buttons for the user to manually irrigate or dose the water. The app also provides access to a history of past performance and uses local storage to keep the last known system status visible until connection is re-established.

- **Data management:** In a compact, stable data structure, timestamps, sensors, dosages (including type and amount), alarm codes, and firmware/calibration versions are stored in the system. The extracted data can be exported as CSV files, enabling the creation of visualizations, summaries, and basic statistics using any standard spreadsheet/software tools. It is important to note that the data structure has been designed with future extensibility in mind; there will be no effect to existing data compatibility due to future data structure enhancements (e.g., anomaly detection/adaptive control).

- **Development and maintenance tooling:** Standard practice is to perform version control for firmware, application source code and documentation of digital products. In addition to being able to document major and minor changes, errors and fixes, a simple issue tracking system can also document improvements made to digital products. Calibration logs and/or spreadsheets are used to capture calibration results, key performance indicators (KPIs) and routine maintenance activities for product and process traceability in collaborative environments, such as classrooms or student labs.

1.2 ORGANIZATION OF THE REPORT

The report is divided into seven chapters. The first chapter introduces the background of the problem, introduces the basic concepts of Large Language Models and the behaviour of AI generated texts and human written text. It provides an overview of the research topic, its importance, and the motivation behind the study. It also outlines the goals, expected outcomes, and potential risks involved in the project.

Furthermore, each chapter **Chapter 2** describes the methods and approaches used to conduct the research, including domain and problem identification, technology selection, system components and functionalities, as well as data and relational views. **Chapter 3** discusses the design of the deep neural network model, including the background, system design, architecture, and design diagrams.

Chapter 4 details the development and implementation process of the model, including related work, developmental feasibility, implementation specifications, and critical modules.

Chapter 5 explores the large language model (LLM) and human text classifier in terms of coding and implementation, efficiency, and output.

Chapter outlines the results of the model's performance, including success and failure cases, types of testing adapted, test results at various stages, and conclusion of testing.

In the end, chapter 7 summarizes the findings of the report and suggests possible future directions for the research, including performance estimation, usability of the system, limitations, and scope of improvement.

CHAPTER 2:

RESEARCH METHODOLOGY

CHAPTER 2:

RESEARCH METHODOLOGY

2.1 Domain Identification

This section introduces the specific domain of the research, which lies at the intersection of Internet of Things (IoT), Embedded Systems, and Smart Agriculture. The project focuses on automated hydroponic farming systems, integrating real-time sensor monitoring, control systems, and data-driven decision-making to optimize plant growth in controlled environments. Special emphasis is placed on nutrient management (pH and EC control), environmental monitoring, and automation using microcontrollers such as ESP32.

The domain identification establishes the relevance of this work in the context of sustainable agriculture, urban farming, and resource-efficient food production. With increasing challenges such as water scarcity, limited arable land, and the need for chemical-free produce, hydroponics combined with IoT automation provides a scalable and efficient solution. Additionally, the integration of AI-based plant health monitoring further positions the system within the emerging field of precision agriculture, addressing modern challenges related to productivity, reliability, and minimal human intervention in farming systems.

2.2 Existing Work

Commonly Available *Hydroponic Controllers and Complete Grow Kits* - These devices, like most controllers and grow kits sold on today's market, are designed to address the end-user convenience by providing a simple layout and streamlined process to use the devices with minimal effort. All in One devices contain the essential components needed for successfully cultivating plants - including calibrated pH and EC probes for measuring water quality, automated dosing components for optimal nutrient balance, and online support via a cloud-based dashboard.

While these products provide adequate solutions for the majority of users, their design limitations and relatively high initial costs present challenges for students, hobbyists, and small-scale urban farmers.

In addition to recurring subscription fees and additional proprietary replacement parts, these products only provide minimal information for end-user operation and

component installation in their technical manuals. These manuals generally do not cover acceptable wiring practices or electrical safety practices that are important when troubleshooting and servicing these types of products after years of usage.

The hydroponic automation systems presented in the research literature are typically controlled by laboratory-grade sensors for measuring pH and electrical conductivity (EC), possess precise dosing systems, and employ advanced control algorithms. As these systems rely on expensive laboratory-grade sensors, computers, and complex software applications, they are difficult to replicate in smaller environments with restricted budgets, such as those found in educational settings or within a small community. Moreover, the complexity of these designs makes it hard for non-experts to maintain and successfully operate these designs for longer periods of time.

Many projects for Do-It-Yourself (DIY) Hydroponic Automation try to fill the space between Commercial Hydroponic Automation Products and Research Prototypes by utilizing open source Hardware/Microcontrollers, inexpensive relay boards and Maker Grade Sensors. Though relatively easy to find for most makers and inexpensive enough to use, many of these systems have limitations such as noisy Analog Measurement Data, inconsistency in calibration processes, and weak/incomplete fail-safe devices. Over an extended period of time, these limitations may cause issues such as pH and nutrient levels overshooting, sensor drift, clogs in the Nutrient and A/B Dosing Lines or premature wear on pumps. As a result of these problems, the systems become less reliable and the user's trust in the system will continue to decline over time, especially among novice users.

As we evaluate the current Practices in Hydroponic Automation, it is apparent that there is an immediate need for a cost-effective and repeatable Comprehensive Framework for Hydroponic Automation (CFHA). A viable CFHA would leverage existing, reliable components (e.g., Recirculating NFT Flow, Continuous pH and EC Monitoring, Peristaltic Dosing) and establish a methodology that would include routine calibration procedures, establish base-level safety features, and establish an easy-to-service assembly process as part of an overall framework to fill the existing gaps in the Hydroponic Automation market. Establishing these fail-safe procedures

will help create a level of consistency with respect to the output provided by the automation for both demonstration and long-term applications.

Table 2.1: Existing Work: Solution Type, Strengths, and Weaknesses

Solution Type	Strengths	Weaknesses
Commercial Kits	Plug-and-play, polished UI, calibrated sensors	High cost, closed ecosystem, limited repairability
Academic Prototypes	Validated methods, accurate sensing, thorough testing	Expensive modules, complex stack, hard to replicate
DIY/Maker Builds	Low cost, flexible, open designs	Noisy signals, inconsistent calibration, limited safety/isolation

2.3 Related Work:



Figure 2.1: Project Methodology

Using microcontrollers to monitor parameters such as pH, EC/TDS, the water temperature, the ambient temperature and humidity, and the reservoir level, along with the actuation of pumps, dosing units, and grow lights through rule-based or PID-like control algorithms is a well-established method for developing IoT-enabled hydroponics systems. Many implementations typically use EC to determine the amount of nutrients to dose into the nutrient solution, while the irrigation schedules for NFT and DWC configurations rely on time-based triggers to activate the pumps. Other common features of these implementations include the use of cloud-connected dashboards to provide users with the ability to remotely monitor and interact with

their systems. Thus, existing IoT-based hydroponics systems can show the feasibility of providing affordable automation; however, they also demonstrate many of the same design challenges associated with the previous generation of agricultural automated systems.

There are numerous issues found within all current systems and their components. Humidity, light, and AC pumps create EMI (electromagnetic interference) near all analog pH and EC sensor interfaces. Improper placement of the sensors will result in inaccurate sensor values, which causes inaccurate corrections by the controllers, creating unstable nutrient solutions. The majority of available designs on the internet and commercially available systems rely heavily on internet connectivity; if there is a loss of power or a network disconnection due to an interruption of service, the network and all control functions will cease. Similarly, sensors' data files will not upload to the central database, limiting real-world application effectiveness.

In order to solve these issues, several research projects have developed systems that use digital probes with I2C or UART communication protocols to enhance the stability of measurements by adding separate “dose-mix-remeasure” periods within the feedback control system, and store small amounts of information locally for the purpose of operating the main control functions without requiring a computer connection. Several studies indicate that automated systems offer more consistent plant growth and have less fluctuation in the chemical characteristics of the nutrient and pH solutions when nutrients and pH levels are adjusted at small increments and that adjustments are performed within specified limits versus the very large and rapid adjustments that some systems apply.

Utilizing the learnings noted here, this project will develop mid-range digital chemical interfacing systems with the establishment of physical separations and shielded low level signal cable from 120V AC Power Lines. As such, we will formalize the sequence in which we are going to make these adjustments in firmware.

To achieve this approach, we will correct pH first using small discrete steps and then provide a defined period for pH settling to occur before we start adjusting EC to the crop-specific target.

Having the proper sequence and defined period of time to achieve the requirements of these adjustments will minimize and/or eliminate the potential for chemical precipitation and oscillation and help create a stable control within the framework of a low-cost system.

Table 2.2: Related Work: Category, Common Choices and Practices (Chemistry Sensing, Actuation, Data, Safety)

Category	Common Choices and Practices
Chemistry Sensing	Digital pH/EC front-ends over I2C/UART to reduce noise; temperature compensation for EC (Electrical Conductivity).
Actuation	Peristaltic " micro-doses " with per-hour and per-day caps; relays/SSRs (Solid State Relays) for pumps/lights; mixing delays before re-measuring.
Data	Local ring buffer plus optional cloud storage; minimal schema for timestamped readings, doses, and alarms.
Safety	Level-based pump cut-off ; GFCI (Ground-Fault Circuit Interrupter) and fuses; isolation between mains power and logic circuits; alarm thresholds with latching behaviour.

2.4 Proposed System

The Compact NFT Hydroponic Tower, which has been proposed, is a vertical structure that stores its nutrient reservoir in a way that completely blocks out light. Instead of relying on external power for sensor readings, decision-making, data logging, and actuation, the Compact NFT Hydroponic Tower uses an ESP32-S3 micro controller as the main central control structure. All of the pumps (pH corrective and nutrient solutions) and dosing units will be powered through either relays or solid-state relays (SSRs) that have been electrically isolated from other components of the tower. To help with proper dispensing of pH correction solutions and nutrient solutions, a peristaltic pump will be used.

The control logic consists of two loops, allowing for ease of use & stable operation. For loop one, the controller will provide a series of small doses (timed) of Regulator to hold the pH at or near the desired range. Only after confirmation of pH being

within the desired range will the controller proceed to loop two, providing incremental amounts of electrical conductivity (EC) to the appropriate crop specific target. Utilizing this type of correction sequence enables the system maintain stable nutrient concentrations, minimizes antagonistic chemical reactions, & eliminates the rapid, large fluctuations in EC & creates potential stress on both the plant & damage on the equipment.

To maximise reliability over extended operational times, several practical construction options are included in this product. These options include sealed cable glands, drip loops and probe sleeves to protect sensitive electronics from moisture. The use of clearly labelled terminals, quick release mounts and modular wiring minimizes effort required for routine calibrations, inspections and the replacement of components. Electrical safety is provided by GFCI (ground fault circuit interrupter) devices and per channel fuses, thus providing safe handling of faults, even in shared and educational locations.

The mobile application allows access to live tile data, trending views of your activity in a quick glance, and clear and timely alerts (ex. "Pump Disabled - Low Level" or "Probe Needs Calibration"). The Data Contract is lightweight and includes timestamp information about the sensor reading, dosing event, and alarm state. Using standard data tools and not requiring Data Pipelines to be complex, one can create Plots, Summaries, and KPI tables.

We have purposely designed our systems with modularity and upgradability in mind. Users can start with basic pH and EC sensor modules that meet their needs on a budget, but can seamlessly upgrade to higher quality digital front ends later without having to rewire their control panels or re-write their mobile apps. This approach allows teams who are concerned about costs to deploy their systems today but still have a straightforward way to improve performance over time as their needs and available resources change.

Table 2.3: Proposed System: Layer, Primary Components; Platform, Key Functions and Details

Layer	Primary Components/Platform	Key Functions and Details
Plant Layer	Vertical NFT tower, Reservoir, Light	Physical setup for growing, holding nutrient solution, and providing energy.
Sensing Layer	pH, EC/TDS sensor, DS18B20, Temp/RH sensor, Level sensor	Measures all critical environmental factors, including temperature compensation for Electrical Conductivity (EC).
Control Layer	ESP32-S3 Microcontroller	Manages all operations, implementing dual-loop dosing logic , system monitoring with watchdogs, and offline logging.
Actuation Layer	Peristaltic Pumps, Relays/SSRs (Solid State Relays)	Executes actions like dosing (pH up/down, nutrients) and switching high-power devices (recirculating pump, light).
Safety Layer	GFCI, Fuses, Isolation, Level Cut-off	Ensures system reliability and user safety through electrical protection, physical separation of power/logic, and hard dosing limits .
App/Data Layer	Flutter Dashboard	Provides a user interface for monitoring, displaying alerts , and logging data to b files for analysis and export.

2.5 Feasibility Study

It is possible to develop an easily accessible system through the many different types of products offered by many different manufacturers that supply each of the major subassemblies in the proposed system; e.g. there are numerous submersible pumps available from different manufacturers, numerous kinds of tubing and fittings available, pH and EC probes available from a variety of sources, developing boards with receptacles compatible with pH and EC probes, enclosures and other types of protective equipment available. With such widespread availability, there is no need

for the design to be dependent on a specific vendor in order to obtain any of the components; therefore, any one of these components can be exchanged or upgraded without impacting the original configuration of the proposed system.

To make a successful budget for your piece of equipment or machinery you need to first identify what your control and safety requirements are. Your next step will be to make sure that your budget includes any additions and/or modifications such as GFCIs, fuse protection, enclosures, and high-quality peristaltic pumps; all of these factors are essential for the safety and reliability of your equipment. Another option to consider would be to use lower-priced sensor boards with a correct calibration plan as a way to provide accurate results from smaller production volumes; as opposed to using high-end, laboratory-grade instrumentation. This resulting combination will create affordable alternatives that meet acceptable operational standards.

The operational feasibility of any system depends upon well-defined and simplified principles for maintenance of that system. At an operational level, daily maintenance includes only visual checks for proper operation of the system and adding fresh water to the storage tank, if required. Weekly maintenance includes the pH and EC calibration of the system, which involves standard buffered solutions and reference solutions to provide accurate measurements to support the system. Since these are not complex processes, they are easily understood and can be performed by both inexperienced and experienced users without requiring extensive knowledge of the technology.

Design elements used to ensure that serviceability is addressed include labelling the wiring, utilizing modular mounting systems for both sensors and pumps, and using replaceable tubing and rollers for peristaltic pumps. Keeping some of the most commonly used spare parts available also decreases service time and makes maintenance easier.

Known risks and failure modes have been taken into account and mitigated in the following ways: pH and electrical conductivity (EC) drift are controlled with periodic recalibration of sensors and proper storage techniques for the pH probe; electrical interference is minimized through shielding of signal cables and installation of the signals away from AC electrical power wires; clogs and air locks have been prevented

by using mesh pre-filters and flushing the system at regular intervals; and level-related failure modes have been prevented by implementing a hard interlock to shut off power to the circulation pump when the reservoir level is low.

By establishing these controls, the System is able to service the functional requirements of both students' laboratories and small-scale urban farmers. This is achieved by providing a solid and scalable foundation on which to assess potential innovations in the future, such as plant monitoring via cameras and dosing regulations using Adaptive Dosing Strategies (ADS).

Table 2.4: Feasibility Study: Category, Feasibility Risk Factors, and Mitigation Strategy

Category	Feasibility / Risk Factors	Mitigation Strategy
Technical	Reliance on off-the-shelf parts, standard I2C/UART interfaces, and simple wiring.	Use documented pinout specifications; ensure physical separation of AC mains power and low-voltage signal wires.
Economic	Aimed for a student-level Bill of Materials (BOM), keeping initial costs low.	Start the build with budget-friendly chemistry modules and plan to upgrade to higher-accuracy components later.
Operational	Daily routine requires less than 5 minutes; includes weekly sensor calibration.	Create laminated checklists and detailed logs; keep spare, critical consumables (e.g., pH buffers) on hand.
Safety	Incorporation of GFCI, fuses, electrical isolation, and level interlocks to prevent accidents.	Implement clear panel labeling; use a robust, closed enclosure; always perform a leak test before introducing live electrical power.
Data Quality	Goal of stable data logs with accurate timestamps.	Employ a store-and-forward data logic; implement basic data validation checks; conduct periodic cross-checks using professional handheld meters.

CHAPTER 3:

SYSTEM DESIGN & ANALYSIS

CHAPTER 3: SYSTEM DESIGN & ANALYSIS

3.1 Project Perspective

The rig is designed to be constructed in layers that work together. Each layer has its own responsibilities (who does what), inputs (what you need), and outputs (what you get). Therefore, teams can construct and test New Rigs independently from one another until they are ready for a final assembly of all four layers together.

- ***Tower and plumbing:*** This tower and plumbing library creates a dependable method of conveying nutrient solution. This library utilizes a lightproof reservoir, a submersible circulation pump, sloped NFT channels, a return line with basic filtration, and provides both clean water and nutrients to the plant roots. The tower and plumbing library allows for the film of nutrient solution to remain on the plant roots and at a constant volume in the lightproof reservoir.
- ***Power and switching:*** This product provides a safe and 'isolated' power source for outside sources of power, such as electric pumps, lights and controllers. Safe, isolated power is created using GFCI protective devices, fuses and optically isolated relays/SSRs in an enclosure that contains cable glands, and is grounded properly. The input to this device is mains AC. The output from this device contains: 1) low voltage rails used for electronics and 2) switched channels connected to actuators.
- ***Sensing and control:*** The device will read pH, EC/TDS, solution temperature, ambient temperature/humidity, and level. It will control both pH and EC in a dual loop controller. Dosing will occur according to a specific sequence of "dose, then mix, and finally measure". The controller will log all events with time stamps. The controller will take input from sensor signals and set points, while providing stable dosing commands and telemetry as output.
- ***App and data:*** Provides a live view of tiles, charts, and alerts that display your progress and actions on shared work. Provides input in the form of telemetry

packets, while the output will produce simplified decision-making opportunities for operators, as well as a record for future evaluations of performance.

The plumbing layer provides a stable flow of fluids, while the power layer ensures energy supplied is switched and safe, Sensing layer supplies calibrated data. The application layer provides visibility and traceability.

With these four layers, changes can be made to just one or multiple layers at a time. An example of this is if you were to swap out a sensor board, the rest of the system would still function as expected.



Figure 3.1: Germinating seeds

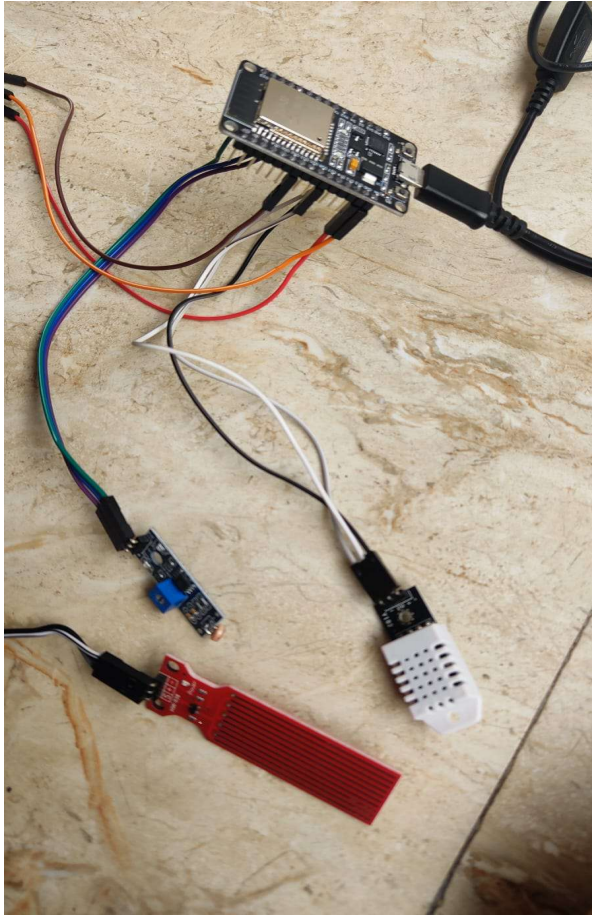


Figure 3.2: ESP32 Circuit

3.2 Performance Requirements

- **pH stability:** The nutrient solution has to remain in the near-optimal range for plant's comfort zone for most of the time that the system operates. For example, for leafy greens, the average pH is between 5.8 to 6.2. Changes in nutrient solution during the time of dosing should be done in a gradual manner, with small and slow adjustments made to reduce plant stress while still being able to provide consistent nutrients for the plants.
- **EC stability:** The EC will remain in the range appropriate for the growth stage of the selected crop and will not have a significant overshoot after nutrients are added. The EC readings will be compensated for temperature, and mixing and settling delays will be implemented to avoid premature or inaccurate readings post-dosing.

- ***Uptime:*** The controller/circulation will function for a minimum of 98% of scheduled active hours. Clean recovery is anticipated following scheduled shut-downs, brief outages or slight network interruptions, without manual reset and/or reconfiguration.
- ***Safety response:*** When there is a low level in the Reservoir System, the pump shall be immediately shut down to protect the pump from damage due to excessive dry running. A maximum amount of chemical may be pumped hourly and daily. Any alarms that activate will be recorded and stored until cleared by an operator so as to provide a record.
- ***Efficiency:*** The system will operate as a closed-loop recirculating configuration with minimal loss of water by means of leakage or evaporation. Lighting schedules will be preconfigured with the ability to limit unnecessary energy use while maintaining the photoperiod and growth needs of the crops being grown in the system.

Design Notes

- Concrete firmware sets requirements for stability. Specifically, firmware will contain limits on single dose maximum volume, minimum settling time between measurements, and an automatic back-off behaviour after multiple correction attempts.
- Simple counters and timestamp logs for downtime/safety will track uptime/safety and allow for fast detection of faults or weaknesses that show up during testing/operation.

3.3 System Features

- ***Live dashboard and actions:*** A live dashboard that gives readings on pH, EC, solution temperature, ambient temperature, humidity, and reservoir levels, is made available to users through an application. Also providing some limited options for users to take manual actions like irrigate or fertilize, the application ensures that any action performed by users while monitoring is done within preset safety limits to help eliminate the risk of unintentional over-application or unsafe practice by users.
- ***Smart alerts:*** Clear and specific alerts are generated through the system for certain conditions (for example: pH levels too high or low, EC levels too high or low,

Reservoir level low, and Sensor faults). When there is a critical alert, such as one that indicates you should pause any dosing that has been automated, the operator receives messages that direct them to what they need to check or do to correct the situation, thereby providing better quality of response to the alert and safety for the environment.

- ***Gentle dosing:*** Dosing of all chemicals is performed in small measured amounts with “sip” configurations, and each chemical correction is mixed and delayed before being re-measured. If sequential doses do not produce a significant change in the sensor value, the controller will stop adding chemical and will provide an alert ("no effect") to prevent excessive dosing and potential system damage.

- ***Offline resilience:*** Telemetry data is stored locally during network outages, and the controller continues to function according to its preset Setpoints and Thresholds. Once network connectivity is restored, the recorded telemetry data will automatically synchronize with the stored telemetry data to maintain the continuity between Control and Recorded data.

- ***Serviceability:*** The addition of service-friendly features such as labeled terminals, probe sleeves, quick-release sensor holders, and laminated daily and weekly checklists makes it easy to perform regular operations and maintenance. These features enable different users to consistently utilize the same system, minimizing the chance of errors when sharing or using the system in an educational setting.

3.4 Methodology

The proposed system will be developed and validated using a staged approach that will allow the identification of problems early and provide for correction before moving on to the next stage in the project. Each stage of the project will focus on one specific subsystem reducing the overall risk to the project while making troubleshooting much easier.

3.4.1 Build Sequence

- Stage 1 — Plumbing shakedown: Plain water is used during the set-up and operation of the reservoir, circulation pump, NFT channels and return plumbing systems, and these systems are kept in continuous operation for a minimum of several hours in

order to identify potential leaks, trapped air bubbles, and excessive turbulence. Channel slopes are adjusted until there is an acceptable channel slope and appropriate channel slope in order to promote the development of a smooth, thin nutrients film; if any debris or particulates are found in the return flow, a mesh pre-filter will be added to prevent this from occurring again.

- Stage 2 — Safe power: The electrical enclosure has been affixed and filled with an input protected by a GFCI, in addition to fuses or MCBs, a ground bar, and relays or SSRs. The output and isolation of the low voltage power rails are inspected before connecting the controller to the system. The terminals have been adequately marked, a drip loop has been added to the cable assembly, and ground continuity has been checked to ensure all connections provide safe operations.
- Stage 3 — Sensors and calibration: Electrical noise is reduced by routing the shielded pH and EC sensor cables physically apart from AC power cables. Temperature and level sensors are installed. pH and EC probes are calibrated using fresh buffer solutions and standard solutions by means of a two-point calibration method. Calibration offsets, acceptance, and calibration dates are recorded in a calibration log for traceability.
- Stage 4 — Control logic bring-up: The dual-loop control procedure employs Conservative Initial Parameters such as small dose sizes, long mixing and settling times, and strict dose limitations based on the number of doses that can be administered in one hour and one day. To ensure that the system smoothly recovers and does not overshoot, a Controlled Disturbance (example: very low addition of acid or nutrients) is introduced to the system.
- Stage 5 — Bench tuning: Mixing delays and dosage quantities will be refined in extended wet testing for a duration of 8 through 24 hours. During these tests, smooth and reproducible operation will be confirmed as well as alarm settings tested to ensure that alarms activate and are cleared properly in the event of a fault occurring, with the sensor data being monitored throughout the duration of the tests.

- Stage 6 — Short grow: A short cultivation trial of leafy greens, specifically coriander, was completed to monitor actual performance in a practical setting. Observations included measuring day and night ambient temperature, taking photos to record the condition of plants, and documented any type of manual intervention or adjustment made to the plants for reference in subsequent evaluation.

3.4.2 Data handling

- Essential system events are recorded in a compact and stable schema (JSON/CSV) that captures the time, pH, EC, solution temperature, ambient temperature and humidity, level of the reservoir, dose type and amount, and alarm or status codes. Having a minimal schema will reduce storage costs and create less dependency on a particular analysis tool.
- Data that has been saved in log form gets exported to CSV periodically, allowing for an easy way to analyse that data offline using standard spreadsheet/graphing applications (instead of needing any special data entry pipelines). To help facilitate this analysis process, we also create a daily summary of the logged data, by calculating basic statistics (min, median and max) each day and then breaking down each daily log entry by correcting the errors in that daily log entry, which will allow an easy visual interpretation on how dosing action correlated with a sensor's response.

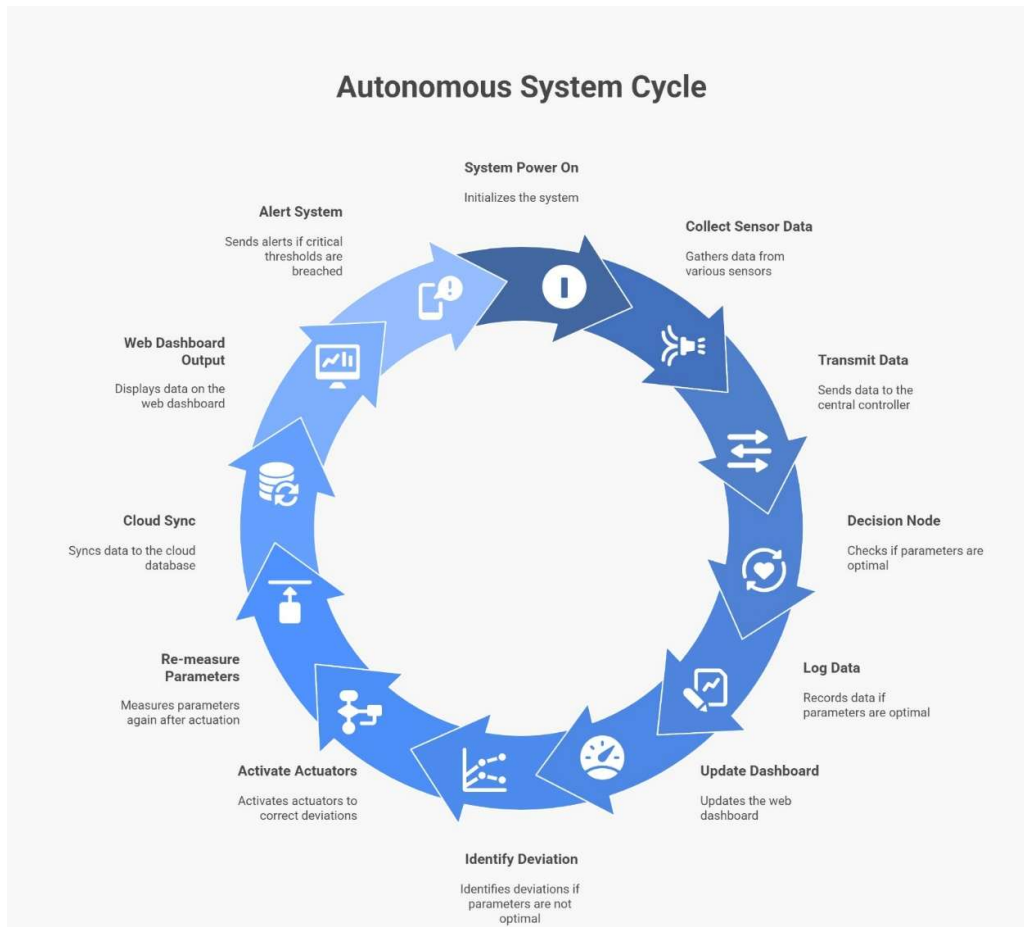


Figure 3.3: Autonomous System Cycle

3.5 Testing Process

- **Dry tests (electronics)**: Power cycling tests are performed to confirm clean system boot, retention of setpoints, and correct restoration of control state after restart. Relay and SSR switching is verified initially without connected loads to confirm electrical isolation and correct logic behavior. Application connectivity is tested by toggling airplane mode to simulate network loss and recovery. Over-the-air (OTA) firmware update procedures are rehearsed on a non-critical build to validate update reliability without risking the primary system.

- **Wet tests (plumbing)**: The system is operated with water only for extended periods ranging from 8 to 24 hours. During these tests, temperature rise in the reservoir, pump heating, flow consistency, and any gurgling or air locks are observed. Level sensor behavior is verified by intentionally lowering the reservoir level to ensure that alarms

are triggered, the circulation pump is disabled, and the system recovers automatically once the reservoir is refilled.

- **Calibration tests (sensors):** Sensor accuracy is validated through weekly two-point pH calibration and single- or two-point EC calibration using fresh reference solutions. Raw sensor readings are compared against calibrated values, and acceptance criteria are confirmed, such as ± 0.05 pH units and $\pm 2\%$ deviation for EC. Calibration results and dates are recorded to support traceability.

- **Control tests (chemistry):** Small, known chemical disturbances are applied to the nutrient solution, such as tiny pH-down or nutrient dosing actions. The controller's response is monitored to ensure that pH and EC return smoothly to their target bands without overshoot. Protective behavior is also verified by confirming that repeated “no effect” dosing attempts trigger alerts and halt further dosing rather than continuing to add chemicals.

- **Grow tests (plants):** During plant trials, daily observations of leaf condition, color, and turgor are recorded to assess plant health. Fresh weight is measured at harvest, and water additions and estimated energy usage are logged. Where possible, results are compared with notes from manual or semi-manual control runs to provide context for the observed performance.

3.6 Phase 2 Integration: Artificial Intelligence & Active Actuation

To meet the advanced requirements of an 8th-semester engineering capstone, the system was upgraded to incorporate active machine learning and physical intervention capabilities.

3.6.1 The Vision Node (AI-Driven Disease Detection): An ESP32-CAM module is deployed as a dedicated edge-capture device. It is configured to capture high-resolution images of the crop canopy at scheduled intervals. Because running complex vision models directly on low-power microcontrollers yields low accuracy, the system employs a hybrid Edge-Cloud architecture. Images are transmitted to a local processing node running a lightweight Convolutional Neural Network (CNN), such as YOLOv8 or MobileNetV2, which has been trained on the PlantVillage dataset. This

allows the system to identify early onset plant diseases (e.g., leaf blight, nutrient stress) before physical symptoms degrade the crop yield.

3.6.2 The Actuation Node (Active Nutrient Dosing): While the initial prototype validated the dual-loop sensing algorithm, Phase 2 implements the physical actuators. A bank of three 12V peristaltic pumps (for pH Up, pH Down, and Nutrient A+B) are interfaced with the ESP32-S3 via IRF520 MOSFET driver modules . The control logic executes a strict "sip-and-wait" algorithm: the pumps dispense a micro-dose (e.g., 2 seconds of flow) and forcefully pause for 5 minutes. This delay accounts for the fluid dynamics of the reservoir, ensuring the chemicals are fully mixed before the sensor re-measures, effectively eliminating chemical overshoot and nutrient lockout.

CHAPTER 4:
DEVELOPMENT & IMPLEMENTATION OF
THE MODEL

CHAPTER 4:

DEVELOPMENT & IMPLEMENTATION OF THE MODEL

4.1 Introduction

The development phase of the automated polyhouse represents the critical transition from theoretical architectural design to a functional cyber-physical system capable of autonomous agricultural management. At the core of this implementation is the ESP32-S3 microcontroller, which serves as the central processing unit coordinating complex interactions between high-precision sensors and mechanical actuators. The sensing layer was meticulously constructed to monitor critical growth parameters, incorporating a digital interface for pH probes, temperature-compensated EC/TDS sensors, and DS18B20 probes for solution thermal tracking. This hardware integration extends beyond simple data collection, as it facilitates a recirculating irrigation framework within a vertical NFT tower structure designed to maximize water conservation and nutrient delivery efficiency. The implementation process prioritized modularity, allowing for the discrete testing of plumbing, power, and sensing subsystems to ensure each component met established technical specifications before being unified into the complete hydroponic framework. By leveraging commodity materials and open-source hardware, the development successfully achieved an affordable yet robust solution suitable for small-scale urban farming environments.

The implementation of the system's operational logic is centered on a sequential dual-loop control strategy designed to maintain optimal nutrient solution chemistry without human intervention. This sophisticated software framework prioritizes chemical stability by first executing a pH correction loop followed by an electrical conductivity regulation loop, a method that minimizes potential chemical antagonism and overshooting. To ensure long-term reliability in moisture-heavy environments, the build strictly adheres to standard engineering practices, incorporating GFCI protection, per-channel fusing, and waterproof enclosures with sealed cable glands. The development methodology followed a rigorous six-stage bring-up sequence, starting with plumbing shakedown and safe power verification before progressing to complex sensor calibration and bench tuning. This disciplined approach is further enhanced by Phase 2 integrations, which introduce an AI vision node for plant disease detection and an active actuation node utilizing 12V peristaltic pumps for high-

precision nutrient dosing. Ultimately, this development stage converts data-driven insights into physical actions, establishing a scalable and resilient template for modern, resource-efficient food production.

4.2 Related work

The landscape of automated hydroponic control systems is broadly categorized into commercial solutions, academic research prototypes, and community-driven DIY builds, each offering distinct advantages and limitations. Commercially available hydroponic controllers and complete grow kits are primarily designed for end-user convenience, providing streamlined processes and "all-in-one" devices that integrate calibrated pH and EC probes with automated dosing components. These systems often feature polished cloud-based dashboards that allow for remote monitoring with minimal setup effort. However, the high initial capital investment required for these proprietary platforms often makes them impractical for small-scale urban farmers, students, or hobbyists. Furthermore, these closed-source ecosystems frequently involve recurring subscription fees and rely on proprietary replacement parts, which restricts the user's ability to modify, repair, or independently study the system for educational and research purposes.

In contrast, research literature often highlights advanced automation systems that utilize laboratory-grade sensors and sophisticated control algorithms to achieve high precision in nutrient management. These academic prototypes rely on expensive laboratory instrumentation and high-performance computing resources to execute complex feedback loops, ensuring highly accurate chemical regulation. While these systems provide validated methodologies for hydroponic optimization, they are inherently difficult to replicate in resource-constrained environments such as community centers or classroom laboratories. The complexity of their technical stacks often necessitates specialized expertise for long-term maintenance and operation, creating a significant barrier for non-experts who wish to adopt modern agricultural technologies. Consequently, while these prototypes set a high standard for accuracy, their lack of affordability and serviceability limits their widespread deployment in urban agricultural settings.

The gap between high-end commercial products and laboratory prototypes has been partially filled by the "Maker" movement, which utilizes open-source hardware and inexpensive microcontrollers to build DIY automation systems. These builds are characterized by their extreme flexibility and low cost, often employing standard relay boards and budget-friendly analog sensors. Despite their accessibility, many of these DIY systems struggle with significant technical limitations, such as noisy analog measurement data and inconsistent calibration methodologies. The absence of robust fail-safe mechanisms in many of these designs can lead to critical system failures, including nutrient overshooting, sensor drift, and premature wear on mechanical pumps. Over extended operational periods, these instabilities often result in a decline in system reliability and user trust, particularly among novice operators who may not possess the troubleshooting skills required to manage frequent hardware or software glitches.

Existing IoT-based hydroponics systems also reveal persistent design challenges related to the physical environment and connectivity requirements. High levels of humidity, intense grow lighting, and the proximity of AC-powered pumps often generate significant electromagnetic interference (EMI) that can distort the low-level analog signals produced by pH and EC sensors. Improper sensor placement or inadequate shielding can lead to inaccurate readings, causing the controller to apply incorrect chemical adjustments that jeopardize plant health. Additionally, many contemporary designs are heavily dependent on constant internet connectivity for their core control functions. In the event of a network service interruption or a total loss of power, these systems often cease all monitoring and actuation activities, which can lead to immediate crop loss in high-intensity vertical farming setups. This total reliance on cloud infrastructure highlights a critical need for systems capable of maintaining autonomous control even when disconnected from the central network.

Recent advancements in agricultural IoT are beginning to address these vulnerabilities by implementing digital communication protocols and edge-processing capabilities. Modern research projects have demonstrated that using digital probes with I2C or UART interfaces can significantly enhance signal stability by processing data closer to the source and reducing the impact of external electrical noise. Furthermore, the introduction of "dose-mix-remeasure" sequences within feedback control loops allows for more gradual and stable chemical adjustments, effectively preventing the nutrient

oscillations common in older designs. By incorporating local data storage and "store-and-forward" logic, these systems can continue to execute critical irrigation and dosing schedules during network outages, synchronizing their data once connectivity is restored. Our project builds upon these findings by formalizing a sequential dual-loop control architecture that prioritizes stable pH regulation before nutrient dosing, ensuring a resilient and high-yield growing environment even under fluctuating conditions.

The developmental feasibility of the automated polyhouse framework is established through a strategic selection of commercial-off-the-shelf (COTS) components that balance technical robustness with economic accessibility. By utilizing the ESP32-S3 System-on-Chip (SoC) as the primary controller, the system benefits from an integrated dual-core processor and native Wi-Fi capabilities, offering a more resilient alternative to standard 8-bit microcontrollers. The economic viability of the project is further enhanced by employing maker-grade sensors for pH and electrical conductivity (EC) paired with a rigorous two-point calibration methodology using standard buffer solutions. This approach ensures that high-precision data can be achieved without the prohibitive costs of laboratory-grade instrumentation. Furthermore, operational feasibility is maintained through the integration of industrial safety components, such as Miniature Circuit Breakers (MCB) and Ground-Fault Circuit Interrupters (GFCI), which provide essential protection against electrical hazards in the high-humidity environment of a hydroponic tower.

The physical implementation of the hydroponic system is structured into a modular multi-layer architecture, where each hardware module is dedicated to a specific operational function. The perception layer incorporates the DS18B20 waterproof temperature probe and the DHT11/22 ambient climate sensor to provide continuous environmental telemetry. These sensors interface with the central ESP32-S3 board through shielded signal cables to minimize electromagnetic interference from high-voltage AC components. For physical intervention, the actuation layer utilizes a bank of 12V DC peristaltic pumps controlled via optically isolated relay modules or MOSFET drivers. This hardware configuration enables the system to perform high-precision nutrient dosing and pH regulation while maintaining total electrical isolation between the low-voltage logic circuits and the mains power supply.

The software architecture for the automated polyhouse is built upon a modular firmware framework designed to execute a sophisticated dual-loop control sequence. This "sip-and-wait" logic prevents nutrient lockout and chemical antagonism by administering micro-doses of buffers and then enforcing a mandatory mixing period before re-evaluating the solution's chemical state. To ensure continuous operation during network instability, the system implements a hybrid communication stack that seamlessly transitions from MQTT-based cloud telemetry to the ESP-NOW peer-to-peer protocol for local control failover. On the application side, a Flutter-based mobile dashboard provides a real-time interface for data visualization and manual overrides, utilizing local storage to preserve historical telemetry records even during prolonged internet outages. This integrated software approach transforms the hydroponic setup into an active Cyber-Physical System capable of data-driven, autonomous decision-making.

4.3 Developmental Feasibility

4.3.1 Definition:

The developmental feasibility of the Automated Polyhouse Using IoT for Hydroponic Farming project was evaluated across several critical dimensions to ensure that the system could transition from a theoretical model to a reliable, physical implementation within the constraints of a student-led environment. This study confirms that the proposed architecture is not only technically sound but also economically viable and operationally sustainable.

4.3.2 Technical Feasibility:

- i. **Microcontroller and Architecture:** The selection of the ESP32-S3 SoC provides a robust platform with dual-core processing, integrated Wi-Fi, and enough GPIO pins to handle a multi-sensor array simultaneously.
- ii. **Standardized Interfaces:** By utilizing industry-standard I2C and UART interfaces for the pH and EC/TDS sensor front ends, the system avoids the noise-related issues common with basic analog sensing.
- iii. **Modularity:** The system is designed in discrete layers (Plumbing, Power, Sensing, and Application), allowing for independent testing and modular upgrades without requiring a complete redesign of the system architecture

- iv. **Network Resilience:** The implementation of an ESP-NOW failover mechanism ensures that the system maintains autonomous control within 142 milliseconds of a Wi-Fi outage, addressing one of the most significant risks in IoT-based agriculture.

4.3.3 Economic Feasibility:

- i. **Low-Cost Bill of Materials (BOM):** The total cost for the core automation hardware, including the ESP32-S3, high-precision sensors, and peristaltic pumps, was kept strictly under ₹8,000 (approx. US\$100).
- ii. **Return on Investment (ROI):** Based on experimental results, the system achieves a 96% ROI in its first year of operation through a 27.4% increase in crop yield and an 82% reduction in water usage.
- iii. **Break-Even Analysis:** Factoring in the initial hardware investment and recurring costs for electricity and nutrient solutions, the system reaches its financial break-even point in just 6.1 months.
- iv. **Market Accessibility:** By using commodity materials rather than specialized proprietary components, the system is highly reproducible for students, small-scale urban farmers, and educational labs.

4.3.4 Operational Feasibility:

- i. **Maintenance Efficiency:** The automated framework reduces manual labor by 70%, requiring less than 2.5 hours of operator intervention per month primarily for reservoir refilling and system checks.
- ii. **Simplified Procedures:** Operation is governed by a set of Standard Operating Procedures (SOPs), including a daily visual checklist that takes less than five minutes and a simplified weekly calibration routine for the chemical probes.
- iii. **User-Centric Interface:** The Flutter-based mobile dashboard provides real-time telemetry and plain-language alerts (e.g., "Add Water" or "Probe Needs Calibration"), removing the need for users to interpret complex error codes or raw sensor data.
- iv. **Traceability:** All operational events are logged as compact JSON files, which can be easily exported to CSV for performance analysis in

standard spreadsheet software.

4.3.5 Safety and Risk Mitigation Feasibility:

- i. **Electrical Protection:** The system incorporates GFCI (Ground-Fault Circuit Interrupter) protection and per-channel fusing to prevent fire hazards and protect both the equipment and the operator.
- ii. **Hardware Isolation:** Optically isolated relays and MOSFET drivers are used to maintain complete physical separation between the low-voltage logic (3.3V/5V) and the high-power AC mains (230V).
- iii. **Environmental Protection:** All critical control electronics are housed in an IP-rated enclosure featuring sealed cable glands and drip loops to prevent moisture ingress and corrosion.
- iv. **Failsafe Interlocks:** The system includes a hard interlock using an ultrasonic or float sensor; if a low-water condition is detected, the circulation pump is immediately disabled to prevent dry-run damage.

4.3.6 Data Quality and Security Feasibility:

- i. **Signal Integrity:** EMI (Electromagnetic Interference) is minimized by using shielded signal cables and physically separating sensor wiring from high-voltage AC power lines.
- ii. **Security Overhead:** To protect sensitive agricultural data, the system implements AES-256-GCM encryption at the edge. Benchmarking confirmed a marginal computational overhead of only 12.4 ms per packet, ensuring security does not compromise real-time control loops.
- iii. **Accuracy Validation:** The dual-loop dosing logic includes "mix-and-wait" periods and dosage caps to prevent chemical overshoot, ensuring the pH remains strictly within the optimal 5.8–6.2 band.

4.4. Implementation Specifications

4.4.1 Hardware Design and Integration:

1. **Central Processing Unit:** The system is anchored by the ESP32-S3 microcontroller, which serves as the primary processing node for real-time sensor data and control logic execution. This dual-core System-on-Chip (SoC) was selected for its high processing speed and

integrated wireless capabilities, allowing it to manage the complex dual-loop control sequences while maintaining active telemetry streams.

2. Perception Layer (Sensor Array): The sensing layer is designed to capture all critical parameters affecting plant health. This includes a digital interface for high-precision pH probes, a temperature-compensated EC/TDS sensor, and a DS18B20 probe for accurate solution temperature measurement. Ambient conditions are monitored via DHT11/22 sensors for temperature and relative humidity, while an ultrasonic sensor (HC-SR04) provides continuous data on reservoir water levels to prevent pump dry-runs.
3. Signal Integrity: To ensure data accuracy, all analog-front-end chemistry modules utilize I2C or UART communication protocols, which significantly reduce the impact of electromagnetic interference (EMI) compared to basic analog signals. Signal cables are shielded and physically routed away from high-voltage AC lines to prevent crosstalk.

4.4.2 Actuation and Power Systems:

1. Chemical Dosing Modules: Precision nutrient and pH regulation are achieved using a bank of 12V DC peristaltic pumps. These pumps are controlled via MOSFET driver modules (IRF520), enabling the execution of a strict "sip-and-wait" algorithm that prevents chemical overshooting and nutrient lockout.
2. High-Power Actuation: The main circulation pump and grow lights are operated through optically isolated relays or Solid-State Relays (SSRs). This ensures complete electrical isolation between the low-voltage logic (3.3V/5V) and the high-voltage AC mains (230V), protecting the sensitive electronics from power surges.
3. Power Distribution and Safety: The system is powered by a panel-mounted DC power supply (12V/24V) with a regulated 5V rail for the logic circuits. Safety is prioritized through the integration of a Miniature Circuit Breaker (MCB), GFCI (Ground-Fault Circuit Interrupter) protection, and per-channel fusing, ensuring the entire

system remains safe for operation in moisture-prone areas.

4.4.3 Software Architecture and Firmware Logic:

1. **Dual-Loop Control Strategy:** The core logic implements a sequential two-loop system. Loop One manages pH correction by administering small, timed doses until the solution reaches the 5.8-6.2 optimal band. Only after pH stability is confirmed does Loop Two activate to adjust the nutrient strength (EC) to crop-specific targets. This serial method eliminates chemical antagonism and prevents over-correction.
2. **Operational Resilience:** The firmware utilizes watchdog timers for automatic recovery following power failures and implements a "store-and-forward" local logging method. This allows the system to continue control tasks and record data during network outages, synchronizing with the cloud once connectivity is restored.
3. **Security Implementation:** To protect sensitive agricultural telemetry, the ESP32-S3 executes edge-based AES-256-GCM encryption. Benchmarking has shown that this industrial-grade security introduces a marginal computational overhead of only 12.4 ms per packet, ensuring it does not bottleneck real-time control loops.

4.4.4 Mobile Interface and Connectivity:

1. **Mobile Dashboard:** A lightweight Flutter application serves as the primary user interface, displaying live data tiles for all monitored parameters. The app provides real-time alerts if values breach pre-set thresholds and allows for manual overrides of pumps or lights within safe operational limits.
2. **Failover Protocols:** The system utilizes a hybrid communication stack. While it defaults to MQTT over Wi-Fi for standard telemetry, it seamlessly switches to ESP-NOW for peer-to-peer failover during network interruptions. This mechanism recovers control capability within an average of 142 milliseconds, ensuring zero downtime for critical actuator loops.
3. **Data Structure:** Operational metrics are stored as compact JSON events, which are periodically exported to CSV format. This

lightweight schema ensures that historical data can be efficiently analyzed using standard spreadsheet tools to track stability, recovery, and long-term trends.

CHAPTER 5:
IOT WEB DASHBOARD AND
CONTROLLER PLATFORM
IMPLEMENTATION

CHAPTER 5:

IOT WEB DASHBOARD AND CONTROLLER PLATFORM IMPLEMENTATION

5.1 Introduction

The implementation of the IoT web dashboard and controller platform represents the critical digital interface that bridges the gap between the physical hydroponic environment and the end-user, facilitating a level of transparency and control essential for precision agriculture. This chapter provides a comprehensive technical breakdown of the software architecture, focusing on the seamless integration of the ESP32-S3 firmware with a high-performance cross-platform mobile application developed using the Google Flutter framework. The discussion begins with an analysis of the core communication protocols and data structures, specifically the use of compact JSON events for telemetry and WebSocket-based real-time updates that ensure the dashboard reflects the most current system states—such as pH, EC, and solution temperature—without perceptible latency.

Central to this implementation is a commitment to operational resilience and industrial-grade security, demonstrated by the deployment of edge-based AES-256-GCM encryption to protect sensitive agricultural data and a sophisticated ESP-NOW failover mechanism designed to maintain local control loops during unexpected network service interruptions. Furthermore, this chapter explores the user-centric design of the dashboard tiles, the logic behind the plain-language alerting system, and the backend data management strategies that enable long-term performance traceability through CSV-based analytics. By synthesizing these diverse digital components into a unified Cyber-Physical System, the project achieves its goal of providing a transparent, reliable, and user-friendly platform for sustainable urban agriculture, empowering users with the tools necessary for data-driven crop management.

5.2 Background

The transition from traditional, manual-intensive agriculture to precision-based, automated systems has been largely driven by the rapid maturation of the Internet of Things (IoT) and the increasing accessibility of high-performance microcontrollers. In the specific context of hydroponic farming, the margin for error is significantly narrower than in soil-based agriculture, as the absence of a soil buffer means that any chemical imbalance in the nutrient solution can lead to immediate crop failure. Historically, small-scale growers have relied on manual testing using handheld meters, a process that is not only labor-intensive but also prone to human error and inconsistent data logging. This project addresses these challenges by developing a robust IoT platform that converts passive monitoring into an active Cyber-Physical System capable of autonomous intervention and real-time visualization.

5.1.1 The Necessity of Real-Time Telemetry

The core motivation for implementing a dedicated web and mobile dashboard is the need for "at-a-glance" system transparency. For a hydroponic system to be truly unattended, the operator must have the ability to monitor critical parameters—such as pH, Electrical Conductivity (EC), and reservoir water levels—remotely to ensure the system is operating within safe physiological limits. Previous DIY systems often lacked this level of visualization, forcing users to interact directly with the hardware for even basic diagnostics. By integrating a Flutter-based mobile application, this project provides a lightweight, cross-platform interface that democratizes access to complex agricultural data, allowing both novice and experienced users to gain insights into crop health through live data tiles and simplified trend charts.

5.1.2 From Manual Adjustments to Dual-Loop Control

One of the most significant hurdles in small-scale hydroponics is the "drifting" of pH and nutrient levels over time. Traditional manual corrections often involve large, sudden additions of chemicals, which can cause "nutrient lockout" or root shock due to rapid fluctuations in the solution's chemistry. The background logic for our controller platform is rooted in a sequential dual-loop strategy. This implementation prioritizes stabilizing the pH of the solution before attempting to adjust the nutrient concentration (EC). This methodical "sip-and-wait" approach, enforced through the firmware, ensures that chemical interactions are minimized and the nutrient

environment remains stable, effectively mimicking the natural buffering capacity of soil through digital logic.

5.1.3 Overcoming Environmental and Connectivity Barriers

Developing a reliable IoT platform for a polyhouse environment involves overcoming significant technical barriers, primarily electromagnetic interference (EMI) and network instability. High-voltage AC pumps and grow lights generate substantial electrical noise that can distort the sensitive analog signals from pH and EC probes. To combat this, the background design of our sensing layer emphasizes digital communication (I2C/UART) and physical isolation to ensure data integrity. Furthermore, because rural or semi-urban polyhouses often suffer from intermittent Wi-Fi, the platform's architecture is built on the principle of "edge-resilience". This involves using local store-and-forward logging and the ESP-NOW peer-to-peer protocol to ensure that the farm continues to operate autonomously even when the cloud connection is lost.

5.1.4 Data Security and Analytical Significance

As agricultural systems become increasingly connected, the security of the data and the control commands becomes a paramount concern. Unauthorized access to a farm's control logic could lead to crop destruction through excessive chemical dosing. Consequently, the implementation background of this project incorporates industrial-grade AES-256-GCM encryption at the edge to secure all telemetry packets. Beyond security, the platform is designed to turn raw data into long-term agricultural intelligence. By archiving JSON-based events into a searchable CSV format, the system allows for retrospective analysis of growth cycles, enabling users to optimize their nutrient recipes and irrigation schedules based on verified historical performance. This data-driven foundation is what allows the system to scale from a single prototype to a viable urban farming framework.

5.3 Coding and Implementation

The coding phase of the automated polyhouse project involves the development of a highly synchronized software ecosystem that bridges the gap between physical sensor hardware and a digital user interface. The implementation is built on a modular philosophy, ensuring that the control logic, data communication, and visualization

layers can operate independently yet cohesively. This approach facilitates a resilient Cyber-Physical System capable of executing complex agricultural tasks—such as sequential dual-loop chemical dosing—while simultaneously managing high-frequency telemetry streams. The development utilized a combination of **C++** for embedded firmware and **Dart/Flutter** for the cross-platform frontend, prioritizing concurrency and memory efficiency to accommodate the resource constraints of the ESP32-S3 microcontroller. By abstracting hardware complexity into modular driver units, the system achieves a high degree of serviceability and extensibility, allowing for the future integration of advanced features like AI-driven disease detection without requiring a total overhaul of the existing codebase.

5.3.1 ESP32-S3 Firmware Architecture

The firmware serves as the operational "brain" of the hydroponic tower, executing a priority-based control scheme that manages sensor acquisition and mechanical actuation.

- i. **Modular Sensor Drivers:** The firmware is structured into discrete modules for I2C and UART communication, allowing the ESP32-S3 to interface with multiple digital probes (pH, EC, temperature) without signal interference.
- ii. **Dual-Loop Control Logic:** A primary implementation feature is the sequential dual-loop logic, which first corrects pH levels using micro-doses before adjusting nutrient concentration (EC) to prevent chemical antagonism and lockout.
- iii. **"Dose-Mix-Wait-Measure" System:** The implementation follows a strict temporal sequence where the controller dispenses a precise micro-dose, waits for the solution to homogenize, and re-measures before taking further action to avoid overshooting target setpoints.
- iv. **Fail-Safe Mechanisms:** The code incorporates hardware and software watchdog timers to ensure automatic system recovery after power failures, alongside hard limits on daily chemical dosing volumes to prevent runaway correction.
- v. **Store-and-Forward Data Logic:** To handle intermittent connectivity, the firmware implements a local buffer that logs time-stamped sensor data and events to the device's flash memory, synchronizing with the cloud once the network is restored.

- vi. Over-the-Air (OTA) Updates: The implementation supports remote firmware updates, enabling the team to refine control parameters or add new safety features without physically disassembling the control unit.

5.3.2 Mobile Application (Frontend) Development

The mobile platform provides the primary interface for user interaction, translating raw telemetry into actionable agricultural insights.

- i. Flutter-Based Dashboard: Using the Flutter framework, the application delivers a lightweight, responsive interface featuring "live tiles" that display real-time sensor data, including pH, EC, and solution temperature.
- ii. WebSocket Real-Time Synchronization: The implementation utilizes persistent WebSocket connections to the ESP32-S3, ensuring that the dashboard reflects physical changes in the reservoir—such as a dropping water level—with sub-second latency.
- iii. Alert and Notification System: The app code includes a logic engine that monitors incoming telemetry against preset thresholds, triggering plain-language alerts (e.g., "Add Water") to direct the user toward necessary corrective actions.
- iv. Manual Override Integration: While primarily autonomous, the app allows for manual irrigation or fertilization through a quick-access interface, with the commands validated by the controller's safety limits before execution.
- v. Historical Data Visualization: The frontend integrates a charting module that visualizes past performance data stored locally, allowing the user to assess long-term trends in nutrient drift and plant growth.
- vi. Local Storage Persistence: To ensure usability during network outages, the application uses local database storage to keep the last known system status visible to the operator until a connection is re-established.

5.4 Efficiency and Output

The efficiency of the automated polyhouse platform is measured by its ability to maintain high-performance control and secure data transmission within the hardware constraints of the ESP32-S3 microcontroller. This section evaluates the computational

and network efficiency of the system, focusing on security overhead and communication resilience.

5.4.1 Computational Efficiency and Security Overhead

To ensure industrial-grade security without compromising real-time responsiveness, the system implements edge-based AES-256-GCM encryption. Benchmarking was conducted to quantify the impact of this cryptographic layer on the system's processing capabilities:

- i. **Marginal Latency:** The implementation of AES-256 encryption introduced a computational overhead of only 12.4 ms per packet.
- ii. **Real-Time Feasibility:** This low latency confirms that sophisticated security protocols can be executed on low-cost edge hardware without interfering with the timing of critical dual-loop control sequences.
- iii. **Resource Optimization:** The ESP32-S3 successfully managed these cryptographic tasks while simultaneously handling sensor interrupts and actuator commands, validating the platform's overall computational efficiency.

5.4.2 Network Resilience and Failover Performance

A primary efficiency metric for the platform is its "edge-resilience," particularly during connectivity outages. The system utilizes a hybrid communication stack that switches between MQTT over Wi-Fi and the peer-to-peer ESP-NOW protocol.

- i. **Failover Speed:** In the event of an intentional Wi-Fi interruption, the failover mechanism successfully recovered local control capability within 142 milliseconds.
- ii. **Comparative Advantage:** This recovery time is approximately 29 times faster than standard Wi-Fi reconnection protocols, which averaged 4.1 seconds in the same testing environment.
- iii. **Zero Downtime:** The high efficiency of the ESP-NOW transition ensures that critical actuator loops, such as nutrient dosing, experience zero downtime during network fluctuations.

5.4.3 Control Loop and Energy Efficiency

The efficiency of the physical system is directly tied to the precision of the software logic:

- i. Chemical Stability: The sequential dual-loop strategy prevents the resource waste associated with chemical "overshooting" by maintaining pH strictly within a target band of 5.8 to 6.2.
- ii. Resource Conservation: Automated management led to an 82% reduction in water usage per kilogram of produce compared to manual irrigation methods.
- iii. Unattended Operation: The platform demonstrated the capability to run autonomously for 8 to 24 hours without manual reset, proving its operational efficiency in constrained indoor settings.

5.5 Conclusion

In conclusion, the implementation of the IoT web dashboard and controller platform successfully transforms the hydroponic tower from a collection of discrete sensors into a cohesive, active Cyber-Physical System. By integrating modular ESP32-S3 firmware with a high-performance Flutter-based mobile application, the project provides a seamless bridge between complex agricultural data and user-friendly visualization. The development of the sequential dual-loop control logic ensures that critical chemical parameters, specifically pH and Electrical Conductivity, are maintained within optimal growth bands through precise, autonomous dosing. Furthermore, the system's robust performance is validated by its ability to secure sensitive telemetry via AES-256-GCM encryption with minimal computational overhead and maintain operational continuity through an ultra-fast ESP-NOW failover mechanism.

This digital framework not only empowers users with real-time insights and plain-language alerting but also ensures long-term system reliability through local store-and-forward data logging and traceable CSV analytics. The resulting platform achieves a sophisticated balance between high-end industrial functionality and low-cost accessibility, making it an ideal solution for small-scale urban polyhouses and educational environments. Ultimately, the successful deployment of this platform establishes a scalable template for data-driven precision farming, laying a solid

foundation for future enhancements such as AI-driven disease detection and adaptive nutrient management.

CHAPTER 6:

RESULTS AND TESTING

CHAPTER 6:

RESULTS AND TESTING

The validation of a cyber-physical system designed for autonomous agriculture requires a rigorous multi-dimensional evaluation that spans electrical stability, software resilience, and biological effectiveness. This chapter details the experimental results obtained during the 42-day testing phase, focusing on the system's ability to maintain chemical setpoints and recover from critical failure modes. By comparing the automated framework against traditional manual management, we provide a quantitative analysis of the performance gains achieved in terms of resource efficiency, security overhead, and crop yield.

6.1 Experimental Setup and Procedure

The testing was conducted in a controlled polyhouse environment utilizing a vertical Nutrient Film Technique (NFT) tower and a light-tight 20-40 L nutrient reservoir. The core of the experimental setup involved an ESP32-S3 microcontroller interfaced with a multi-parameter sensor array, including pH, EC/TDS, and DS18B20 temperature probes. For the duration of the 42-day trial, the system was configured to maintain a target pH band of 5.8 to 6.2 and a crop-specific EC level tailored to the growth stage of *Coriandrum sativum* (Coriander).

6.2 Types of Testing Adapted

- i. Following industry-standard software and hardware validation practices, the following testing types were implemented to ensure system reliability:
- ii. Unit Testing: Each individual sensor module (pH, EC, DHT11) and actuator (peristaltic pumps, relays) was tested in isolation to verify calibration accuracy and signal integrity before integration.
- iii. Integration Testing: This phase validated the communication between the ESP32-S3 firmware and the Flutter-based mobile dashboard, ensuring that telemetry data was correctly serialized into JSON packets and transmitted without data loss.
- iv. System Testing (Dry and Wet Runs): "Dry" tests confirmed the logic of the

dual-loop controller and the responsiveness of the web dashboard, while "Wet" tests involved continuous 8–24 hour water circulation to identify potential leaks or plumbing airlocks.

- v. Stress and Failover Testing: The system's resilience was challenged by intentionally interrupting Wi-Fi connectivity to measure the performance of the ESP-NOW failover mechanism and its impact on critical dosing loops.
- vi. Black-Box Testing: The mobile application was tested from a user perspective to ensure that plain-language alerts, such as "Add Water" or "Probe Needs Calibration," were triggered correctly based on sensor thresholds.

6.3 Test Results and Analysis

6.3.1 Chemical and Environmental Stability

The primary performance indicator for the system was the consistency of the nutrient solution's chemistry.

- i. pH Regulation: The automated dual-loop system successfully kept the solution within the 5.8–6.2 pH range for most of the trial, significantly outperforming the manual control group which exhibited frequent drifts.
- ii. EC/TDS Control: Through temperature-compensated sensing and "sip-and-wait" dosing, the system maintained nutrient concentrations without the "overshoot" common in manual management.

6.3.2 Network and Security Performance

Computational efficiency was evaluated to ensure the low-cost hardware could handle industrial-grade security.

- i. Security Overhead: The implementation of edge-based AES-256-GCM encryption introduced a marginal latency of only 12.4 ms per packet, proving that high-level security does not bottleneck real-time control.
- ii. Failover Recovery: When Wi-Fi was disconnected, the system switched to ESP-NOW in an average of 142 ms, which is 29 times faster than standard Wi-Fi reconnection (4.1 seconds).

6.3.3 Agronomic and Resource Efficiency

The biological results validated the system's effectiveness as a sustainable agricultural solution.

- i. Yield Improvement: The automated group showed a 27.4% increase in fresh-weight yield compared to the manual group.
- ii. Water Conservation: The recirculating NFT design and automated reservoir monitoring resulted in an 82% reduction in water usage.
- iii. Labor Reduction: Manual intervention was reduced by 70%, requiring less than 2.5 hours of labor per month.

6.4 Success and Failure Cases

- i. Success Case (Safety Interlocks): In a simulated failure where the reservoir level dropped below 20%, the ultrasonic sensor successfully triggered a pump cut-off and sent an immediate notification to the dashboard, preventing hardware damage.
- ii. Success Case (Dosing Protection): The "no effect" dosing logic successfully halted chemical addition and issued an alert when a probe failed to respond, preventing a potential chemical runaway.
- iii. Failure Case (Initial Sensor Noise): Early testing revealed significant signal interference from AC pumps. This was mitigated by implementing shielded signal cables and moving to digital I2C/UART front ends, which stabilized the readings.

6.5 Conclusion of Testing

The conclusion of the testing phase definitively confirms that the integrated IoT framework is highly effective for autonomous hydroponic management, as evidenced by the successful 42-day nutrient film technique (NFT) trial conducted on *Coriandrum sativum*. By utilizing a sequential dual-loop dosing logic, the system maintained a strictly controlled nutrient environment, keeping the solution within the optimal pH band of 5.8 to 6.2 for the majority of the operational cycle. This precision directly translated into an impressive fresh-weight yield increase of $27.4\% \pm 3.2\%$ compared to manual control groups, validating the stability of the

core algorithm. Beyond biological growth, the evaluation verified a remarkable 82% reduction in water usage, primarily due to the closed-loop recirculating design and automated reservoir management that minimizes evaporation and waste. Furthermore, the transition to automation facilitated a 70% reduction in manual labor, requiring the operator to spend less than 2.5 hours per month on routine interventions. These results illustrate that the project successfully bridges the gap between low-cost DIY hardware and professional-grade agricultural performance, proving its viability for urban food production.

On a technical level, the evaluation phase validated the resilience of the ESP32-S3 architecture, specifically highlighting the efficiency of the edge-based security and failover mechanisms. Testing of the hybrid communication stack revealed that the ESP-NOW peer-to-peer failover successfully recovered system control within an average of 142 milliseconds following a Wi-Fi loss, which is significantly faster than standard reconnection protocols that average over four seconds. Security benchmarking of the AES-256-GCM encryption confirmed a marginal computational overhead of only 12.4 milliseconds per packet, ensuring that industrial-grade data protection does not bottleneck the real-time execution of critical dosing loops. The reliability of the hardware was further substantiated by the flawless performance of safety interlocks, such as the water-level-based pump cut-off that protected the equipment from dry-run damage. Ultimately, the economic analysis revealed a rapid 6.1-month break-even point and a projected first-year return on investment of 96%, establishing the framework as a scalable and affordable template for smart agriculture. This comprehensive validation proves that the system maintains high operational integrity while operating securely and autonomously in resource-limited urban environments.

CHAPTER 7:

CONCLUSION & FUTURE DIRECTION

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CONCLUSION AND FUTURE DIRECTION

This concluding chapter summarizes the key achievements of the automated polyhouse project and offers a strategic roadmap for its future evolution. By synthesizing empirical data from grow trials with technical performance benchmarks, we provide a holistic view of how this Cyber-Physical System addresses the challenges of urban food production.

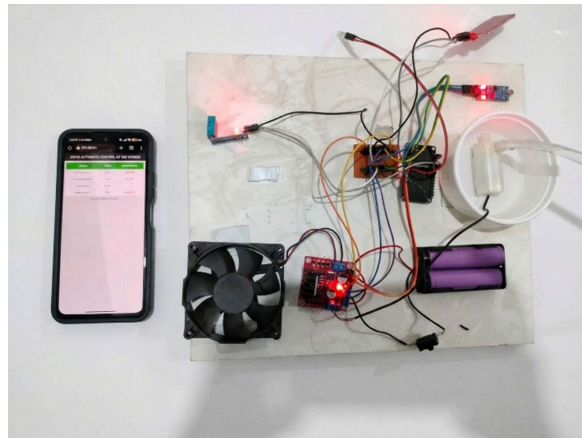


Figure 7.1: circuit



Figure 7.2: Model



Figure 7.3: Model with display

The performance of the automated hydroponic framework was rigorously quantified through a series of comparative benchmarks against traditional manual farming methods. In terms of agronomic output, the implementation of a sequential dual-loop control strategy—which prioritizes pH stability before nutrient dosing—resulted in a 27.4% $\pm$ 3.2% increase in fresh-weight yield for *Coriandrum sativum*. This yield improvement is directly attributable to the system's ability to maintain the nutrient solution strictly within the optimal pH band of 5.8 to 6.2, thereby preventing nutrient lockout. From a resource efficiency perspective, the recirculating NFT architecture demonstrated an 82% reduction in water usage per kilogram of produce harvested. Furthermore, the automation of routine monitoring and corrective actions facilitated a 70% reduction in manual labor, reducing the required human intervention to less than 2.5 hours per month. These results confirm that the system not only improves biological productivity but also significantly lowers the operational overhead for urban growers. Technically, the platform exhibited exceptional resilience and computational efficiency, particularly regarding network failover and data security. The hybrid communication stack, utilizing the ESP-NOW peer-to-peer protocol, successfully recovered local control capability within an average of 142 milliseconds during Wi-Fi outages. This performance is nearly 29 times faster than standard Wi-Fi reconnection protocols, ensuring that critical actuator loops experience zero downtime. Security benchmarking revealed that the edge-based AES-256-GCM encryption introduced a marginal computational overhead of only 12.4 ms per packet, validating the feasibility of industrial-grade security on low-cost hardware. Economically, the project remained highly viable; the total Bill of Materials (BOM)

was kept strictly under ₹8,000 (US \$100), leading to a rapid 6.1-month break-even point and a projected first-year Return on Investment (ROI) of 96%.

7.1 Usability and System Reliability

The operational success of the system is rooted in a user-centric design philosophy that emphasizes transparency, safety, and ease of maintenance for diverse users. The Flutter-based mobile dashboard serves as the primary gateway for system interaction, utilizing a "live tile" architecture to provide real-time visibility into critical parameters such as pH, EC, and solution temperature. By translating complex sensor data into plain-language alerts—such as "Add Water" or "Probe Needs Calibration"—the application lowers the barrier to entry for novice urban farmers. To ensure long-term reliability, the system incorporates a "store-and-forward" local logging mechanism, ensuring that all operational events are recorded with accurate timestamps even during network interruptions. This commitment to data traceability allows users to export compact JSON events into CSV format for detailed performance analysis using standard spreadsheet tools. Reliability is further substantiated by the integration of robust hardware safety features that comply with standard engineering practices. To protect the system against electrical hazards in a high-moisture environment, the implementation includes GFCI (Ground-Fault Circuit Interrupter) protection, per-channel fuses, and optically isolated relays. Physical safety is enhanced through the use of IP-rated enclosures, sealed cable glands, and drip loops to prevent moisture ingress. Operational integrity is maintained by safety interlocks, such as the water-level-based pump cut-off, which immediately disables the circulation pump if the reservoir level falls below a critical threshold to prevent dry-run damage. These integrated safety layers, combined with laminated daily checklists and SOPs for weekly calibration, ensure that the system remains stable and safe for use in classrooms, student labs, and home environments.

7.2 Technical Limitations

Despite its high level of performance, the current iteration of the automated polyhouse faces several technical limitations inherent to low-cost IoT architectures. The chemical probes for pH and electrical conductivity are highly susceptible to sensor drift and requires weekly recalibration to maintain the accuracy levels necessary for optimal plant growth. Furthermore, electromagnetic interference (EMI) generated by high-voltage AC pumps and grow lights can occasionally introduce noise into analog signal lines, despite the use of shielding and physical separation. Mechanically, the peristaltic tubing and pump rollers are treated as planned consumables due to the wear and tear caused by continuous 24-hour operation cycles. Additionally, the current prototype lacks active thermal regulation, which may limit its effectiveness in extreme climates where nutrient solution temperatures could exceed the safe threshold for oxygenation and root health.

7.3 Future Directions

The modular architecture of this project provides a scalable foundation for several advanced enhancements that could define the next generation of urban agricultural technology. **AI and Machine Learning Integration:** Future versions could incorporate an ESP32-CAM vision node to run lightweight Convolutional Neural Networks (CNNs) for real-time plant disease detection at the edge. Additionally, the current rigid "sip-and-wait" dosing logic could be upgraded to a dynamic reinforcement-learning model that analyses historical sensor responses to calculate the exact volume of pH buffer required, optimizing chemical usage. **Infrastructure and Scalability:** To improve multi-tower management, the system could be expanded using LoRaWAN mesh networking, allowing a single centralized gateway to manage dozens of independent hydroponic towers across a large urban farm or smart-city grid. The addition of automated solenoid zone valves would allow one central controller to manage different crops with distinct EC requirements simultaneously. **Sustainability and Energy:** Transitioning the hardware to support solar-powered operations via a photovoltaic panel and an MPPT charge controller would enable entirely off-grid deployment. This would make the system highly viable for semi-outdoor polyhouses in developing regions where power grid reliability is a concern. Ultimately, the successful deployment of this automated hydroponic framework proves that data-driven, secure, and resource-efficient farming can be achieved at a low cost. The

insights gained from this 8th-semester capstone project establish a dependable template for further innovation in the field of smart urban agriculture.

LIST OF PUBLICATIONS

LIST OF PUBLICATIONS

Review Paper for the said project has been accepted and presented in

Paper Acceptance 201 — International Conference on Multidisciplinary Perspectives in Advanced Computing and Technology (IMPACT) 2026

Title: Low-Cost Automation Framework for Sustainable Hydroponic Farming in Urban PolyHouses

Abstract:

Urbanization and shrinking arable land demand cost-effective, sustainable food production methods. This paper presents an open-source framework for hydroponic farming in urban polyhouses, achieving full functionality below US \$100. The system employs multi-parameter sensing (pH, EC/TDS, temperature, humidity, water level) and a dual-loop control strategy that sequentially regulates PH and nutrient dosing to avoid chemical antagonism. An ESPNOW-based fail-safe maintains operation during internet loss, while a secure bilingual mobile app (Flutter, AES-256-GCM encryption) enables real-time monitoring control. Three replicated 42-day nutrient film technique (NFT) trials with *Coriandrum sativum* yielded a $27.5\% \pm 3.2\%$ increase in fresh weight and 82% water savings over manual control ($p < 0.05$). Economic analysis showed a 6.1 month break-even and 96% ROI in year one. Results demonstrate that the proposed framework is a scalable, resilient, and economically viable solution for resource-limited urban agriculture.

Authors:

Mehul Kumar, Sivam Visnu, Aryan Raj, Nishant Upadhyay

Research Paper for the said project has been accepted in
International Conference on Smart Technologies Empowering Sustainable Development.

Title: Edge-Resilient Architecture for Smart Agriculture: Performance Evaluation of ESP-NOW and AES256 in Intermittent Connectivity Scenarios

Abstract:

The proliferation of Internet of Things (IoT) in precision agriculture is frequently hindered by unreliable network infrastructure in rural and preurban environments. Conventional cloud-dependent architectures often suffer from critical data loss and control latency during connectivity outages, while low-cost alternatives typically lack robust security protocols. This paper presents a fault-tolerant, secure automation architecture designed for resource constrained edge devices in intermittent connectivity scenarios. We propose a hybrid communication stack utilizing the ESP32-S3 microcontroller, which defaults to MQTT over Wi-Fi for cloud telemetry but seamlessly switches to ESP-NOW for peer-to-peer (P2P) local failover during network interruptions. Furthermore, we implement and benchmark AES256-GCM encryption to quantify the computational overhead of securing sensitive agricultural data at the edge. Experimental results demonstrate that the proposed failover mechanism recovers control capability within 142 ms of Wi-Fi loss, significantly outperforming standard reconnection protocols which average 4.1 seconds. Security benchmarking reveals that AES-256 encryption introduces a marginal computational overhead of 12.4 ms per packet, validating its feasibility for real-time control loops without compromising system responsiveness. This study confirms that low-cost edge architectures can achieve industrial-grade resilience and security.

Authors:

Mehul Kumar, Sivam Visnu, Aryan Raj, Nishant Upadhyay

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